Quantum Mechanics

(reference guide)

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What one fool can do, another can. (Ancient Simian Proverb.)

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0.1 Preface

The *cover* is a figure of the infinity symbol. These notes takes broad studying to comprehend.

Abstract Mathematics(models of inductive reasoning), is availed to avoid nonsensical answers, and bogus non-differentiable functions. This is the task for the human mind; to see this truth, and preserve it. It's accepted that the Angular Momentum of a closed system is conserved(hence production of particles by conversion of gravitational energy into matter energy, conditioned by Biological factors). The origin of the universe(a Physical picture of a polarizable space, and characterized as young by $\frac{\epsilon}{\epsilon_0}$ 1, with an energy density when expressed in Newtonian units), is not a reproducible event, and cannot be duplicated in the laboratory. In trying to eliminate the Riemann curvature tensor, there's also the Ricci curvature tensor, assuming the photon travels without a shift in energy or frequency(red shift). There are formidable problems facing String Theory: it has no predictive power, it's a highly sophisticated theoretical laboratory(rather than experimental), it's defined at the Planck energy(10^{19} GeV), as any Physical theory that you can possibly come up with, it cannot be tested with the present technology, and it's tightly constrained as much as the graviton emerges as a massless state with spin-2. Space-time is beyond being verified experimentally, and of the vacua states, only one is needed to be isolated, and the Cosmological constant is very close to 0. However, in this theory, when there's spontaneous symmetry breaking, one does not know how to keep the Cosmological constant 0, perhaps a solution will be found to isolate the one vacuum.

No space/time for boredom! Have fun, and ask a Physics lab fundamental questions.

The letters in bold transform like vectors!, and the entire notebook parses on TeX Live 2023/Debian kpathsea version 6.3.5, jupyter notebook/lab, and ipynb viewer(Android).

These are notes/references, few functional code!, and compiling.

An electron plays a central role, and can be assumed to move in a spherically symmetric field, where all binding energies are negative. These are advanced core notes(reference sheet) in Quantum Mechanics(describing how particles behave like waves - the state of a given system[internal structure], and its' probabilities), and some equations of Mathematical Physics[some theorems are mentioned to emphasize the concept of unitarity]. The key important formulas are the Schrödinger equation, Uncertainty Principle, energy levels of the hydrogen atom, harmonic oscillator energy, and the Hamiltonian energy(of which describes how a system evolves in time). Included, are physical quantities, long expressions, Gauge Theory, important systems(like the Harmonic Oscillator, the energy levels of the Hydrogen atom), Mathematical tools, important inequalities, and some advanced notes on Quantum Field Theory. The multiplication by i makes it possible to carry over to QM(Quantum Mechanics)...this is one of the most striking features of Classical Mechanics(CM)-namely, the correspondence between observables, and one-parameter groups of symmetries. We hereby have a Mathematical model capable in principle, of supplying the answer to any Physical question we might ask. The fact that the eigenvalues check with the observed spectrum of the hydrogen atom, when one takes $\mathbf{K} = \frac{\hbar}{2\pi}$, which provides a confirmation of the identity of $\hbar$, with $\frac{\hbar}{2\pi}$. Your objective, dear reader, is to show in theorems of analysis, providing asymptotic formulas for the eigenvalue distribution of differential operators. Don't try to read these notes linearly, instead, come back here as a formula reference.

0.2 List of Figures in Chapters

1. Binary Entropy Diagram
2. Vector pointing towards the Sun
3. Quantum Harmonic Oscillator
4. Anharmonic vs simple Harmonic Oscillator
5. Relativistic mass increase
6. Ground state search
7. Sampled wave function distribution

8. Breit-Wigner distribution

0.3 Diagnostic

(a) Show that $y = 2x^2$ is a solution to the differential equation: $\frac{1}{2}x^2y'' - xy' + y = 0$,

(b) If $y_1(t) = e^{3t}$, and $y_2(t) = e^{-3t}$ are solutions to $y'' - 9y = 0$, what is the general solution? Note that y_1 and y_2 are NOT constant multiples of one another, so they are linearly independent. Thus the general solution in the differential equation is
y(t) = $c_1e^{3t} + c_2e^{-3t}$.

(c) If $y^{''} - 2y^{'} + 5y$= 0, has the associated characteristic equation $\lambda^2 - 2\lambda + 5 = 0$,
by the quadratic formula, the roots of the characteristic equation are $1{\pm}2i$. Therefore, the general solution to this differential equation is y(x) = $e^x(c_1cos2x + c_2sin2x)$.

(d) Knowing when to apply the **Product and Chain Rules, Total and Partial Derivatives**.

NOTE: $\frac{\hbar}{i}$ = -i$\hbar$ (Multiplying by -i is the same as dividing by i.)

PROCEED

Note

Elementary logic functions(Boolean gates), or Disjuctive Normal Forms, Computation complexities, Algorithms, and Quantum circuits, are not part of this material.

• *Schrödinger Equation: General linear homgeneous solution(from translation operators), to the time-dependent Schrödinger Equation* $\longrightarrow$

$$\psi(x,t) = \sum_{n=1}^{\infty} C_n\psi(x)e^{-\frac{iEt}{\hbar}}, \quad \textbf{\textit{t = 0}} \tag{1}$$

• *Integral form, time-dependent* $\longrightarrow$

$$\psi(x,t) = \int \mathbf{K}[x,t:q_0,t_0]\psi(q_0,t_0)dx_0, \textbf{\textit{where K is the kernel or propagator}}. \tag{2}$$

Schrödinger Time Independent Equation

$$\boxed{E\psi_0(x) = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi_0(x)} \tag{3}$$

• *The identity Zero: Zero has no multiplicative inverse!!*

• *Informational Warning: All descriptions here are in the literature, and not original. DO NOT attempt to frame a physical picture!!, or take Relativity into sufficient account. Quantum Field Theory follows once these principles are mastered.*

• *On Rational numbers: If (a) $\neq$ 0, there is no rational number($\mathbb{Q}$), where (b) $\times$ 0 = (a), can be fulfilled.*

Chapter 1 Familiar, recognized, and understood equations

If $\frac{d\alpha}{dt} = 0 \, \forall \, t$*, then* α *is constant*

$(f, \phi,) = \lim_{\alpha \to 0}(f, \phi_n) = 0$

$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$

$e^{i\mathbf{a}\cdot\mathbf{X}}e^{1\mathbf{b}\cdot\mathbf{P}}\phi_0$ *is also a minimum uncertainty state*

$a^{\dagger}|n\rangle = \sqrt{n+1}|n+1\rangle$

$$\hat{j}_{\pm} \mid j, m\rangle = \hbar\sqrt{j(j+1) - m \mid m \pm 1)} \mid j, m \pm 1\rangle \tag{1.1}$$

$\Longrightarrow$

Angular-momentum identity/Discrete orientation, with Quantum-number. *(In units of* $\frac{\hbar}{2\pi}$*).*

$A(t) = e^{+\frac{iHt}{\hbar}}\mathbf{A}e^{-\frac{iHt}{\hbar}}$

$\hat{\mathbf{H}} = p\hat{q} - \mathbf{L}$

$ct' = \gamma(ct - \beta x)$

$E = (\mathbf{p^2}c^2 + m^2c^2)^{\frac{1}{2}}$

$\mathbf{E} = -\frac{\dot{\mathbf{A}}}{c} - \nabla\varphi$

$\forall F \in HL^2(\mathbb{C}, \mu_\hbar)$*, we have:*

$$\|T_{\mathbf{a}}F\|^2L^2(\mathbb{C}^n, \mu_\hbar) = (\pi\hbar)^{-n}\int \mathbb{C}_n e^{-\hbar|\mathbf{a}|^2}\mathbf{e}^{-2\mathbb{R}(\bar{\mathbf{a}}\cdot\mathbf{a})}|\mathbf{F}(\mathbf{z}+\hbar\mathbf{a})|^2\mathbf{e}^{-\frac{|\mathbf{x}|^2}{\hbar}}dx = (\pi\hbar)^{-n}\int \mathbb{C}_n e^{-\frac{|\mathbf{z}+\hbar|^2}{\hbar}}|\mathbf{F}(\mathbf{z}+\hbar\mathbf{a})|^2dz \tag{1.2}$$

$= \|\mathbf{F}\|^2 \; L^2(\mathbb{C}^{\ltimes}, \mu\hbar)$

$\prod(e^X) = e^{\pi(X)} = \sum_{x=0}^{\infty}\frac{\pi X^m}{m!}$ *(Each element is a product of Lie algebra elements. Its power series is finite dimensional, and closed.)*

The Hilbert-Schmidt-norm of a matrix $X \in M_n(\mathbb{C}) \longrightarrow$

$$\|X\|_{HS}^2 = \sum_{j,k=1}^{n}|X_{jk}|^2 \tag{1.3}$$

Cauchy-Schwarz inequality

$$\|XY\|_{HS} \leq \|X\|_{HS}\|Y\|_{HS}\forall X, Y \in M_n(\mathbb{C}) \tag{1.4}$$

SO(3) $\longrightarrow$

$$\sigma_\ell = \pi'_\ell \circ \phi^{-1} = \pi_\ell \tag{1.5}$$

$\hat{\otimes} \longrightarrow$ *Hilbert tensor product*

Feynmann Path-Integral $\longrightarrow$

$$e^{-\frac{it\hat{H}}{\hbar}}\psi(X_0) = C\int_{paths\ with X(0)=X_0} exp\left\{\frac{i}{\hbar}(x(\cdot),0,t)\right\}\psi(x(t))Dx \tag{1.6}$$

Gauge-transformation $\longrightarrow$

$$U_\gamma Q^1_{pre}(f)U_\gamma^{-1} = Q^2{}_{pre}(f), \quad where(U_\gamma) \text{is the map, and "pre" is for pre-quantization.} \tag{1.7}$$

$a' = a - \nabla\psi \qquad \phi' = \phi + \frac{1}{c}\,\frac{\partial\psi}{\partial t}$

This was David Bohms' view of the character of Electromagnetic potentials.

$$\psi_n(r,\phi) = f(r)e^{-in\phi}, \quad \text{where n is an integer, and f is an arbitrary holomorphic function } on(C), with \int_0^\infty \mid f(r)\mid^2 rdr < \infty \tag{1.8}$$

Sympletic-potential $\longrightarrow \theta = \frac{1}{2}$*(p dx - x dp)* = $\frac{1}{2m\omega}$*(p dy - y dp)*

Fourier-transform($\mathcal{F}$) $\longrightarrow$ *A constant multiple of a unitary map* $\longrightarrow \psi(p)$ = - $\int_{\mathbb{R}} \phi(x) e^{-\frac{ixp}{\hbar}}\ dx\ dp$

$C_c^\infty(\mathbb{R}^\kappa) \longrightarrow$ *Space of smooth, compactly supported functions*

1.1 Common nomenclature

1.1.0.0.1 Nomenclecture

1. Written in polar coordinates, ρ and θ are in the unit circle; When $\rho < 1$, we have the series u = $\sum_{\nu=1}^{\infty} a_\nu \rho^\nu \cos\nu\theta$
2. $\pm$ means $-1^{|r|}$
3. ∇ - nabla Differential operator in ordinary 3D space
4. ∇h - Gradient of scalar h
5. $\nabla \cdot \mathbf{A}$ - Divergence of a vector
6. ∇^2 - Laplacian operator in ordinary 3D space spacetime $\longrightarrow \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
7. $\square$ - nabla operator in 3D Minskowski spacetime
8. $\square^2$ - d'Alembertian operator in Minskowski 4D spacetime
9. $\overline{V}$ = $(V^\perp)^\perp$ is the closure of V
10. $g_{i,j}, g^{i,j}$ - Covariant, contraviant metric tensor of 3D space, or its components
11. ϵ_0 - Permitivity of free space
12. μ_0 - Permeability of free space
13. ρ, ϕ, z - Cylindrical corodinates in 3D space
14. σ - Spacetime interval in 4D
15. ϕ - Electromagnetic scalar potential
16. ω - Angular-frequency/speed

1.1.0.0.2 Structure

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The content of the book is based on spaced structuring, and iterative deepening.
```

I would like to appreciate LaTeX , and Overleaf, for making ""learning coding" possible, for industry standards.

Chapter 2 Physical Ideas that captivate

2.1 Characteristics of describing Energy

There is NO electronic ink for philosophical arguments! i.e. local realism does not exist, how long will $\mathbb{Z}$ remain a bachelor, cold fusion, climate change, singularities, why wave-functions overlap, determinism, hidden variables(proof that subquantum level don't exist), why periodicity of quantum numbers lead to periodicity of chemical elements, spin-flip transitions(oscillating magnetic monopoles/distance transverse magnetic $\mathbf{B}$ produced by an oscillating magnetic dipole, etc.... Ultra-cold atoms(lock-stepped with the same spin), have been realized in the laboratory(Bose-Einstein condensate[BEC]), through observing the center of mass system(a trick) of wave-like properties of atoms, so that the Pauli exclusion principle is NOT violated. This technology, although limited(interaction with the external enviroment depolarizes particles) has applications in quantum computers(cryptography - quantum key distribution ; Refer to figure 2.1), and is realized only for very low temperature Kelvin scales, not for classical computers. Even the speed of light (c) - has been stopped!, using slow light pulses. The wave-particle duality(using a super-computer) was proved by an experiment in 1974 by John Clauser(concludes that the wave-function collapse is an objective $\mathbb{R}$ process - not a physical process, and not relativistic invariant - agreed). The idea of measurement of the wave-function with any apparatus(alters the azimuthal quantum number/angular momentum of the orbital plane - resulting to a breakup of eigenvalues into a one-dimensional representation cylindrical symmetry of 2j + 1, from spherical symmetry of j. Why, currently no answer/solutions...maybe its' even a wrong question. Considering Faradays' Law, knowing the Fourier transform, invoke a linear and orthogonal tranformation to Stokes' theorem, generalizing how the electric field remains constant, the magnetic field too (curl$\mathbf{E}$) = - ($\frac{\partial \mathbf{B}}{\partial t}$), with $\iint_s \mathbf{F} dS = \int_c \mathbf{F} dt$, where C is a closed curve, and $\mathbf{F}$ is a vector field defined in $\mathbb{C}$. It can be calculated by making the electric field constant, and varying the magnetic field e.g. $\mathbf{B}(t) = \langle tx, ty, -2tz \rangle$, $0 < t < \infty$. Note that in general, the z component of angular momentum is $\hbar$m, with probability $\nVdash$, and (m) is the state of the magnetic quantum number. This analysis is ficticiously true, and is based on accepting the theory of of complex numbers, Maxwells' equations, Fourier transform, homogeneous linear differential equations, the double-slit experiment(the results on the screen is a transmitted split beam intensity profile), Stern-Gerlach experiment(represents an entangled magnetic moment of Silver atoms), and Schrödingers' equation. There are no visible sources of some forms though, they're(strange and arbitrary) schematic representation of Physical ideas(Force, Momentum, Energy, Mass, Work, Acceleration. Length Of Interval, Time, and Velocity)...are all approximations! The simultaneous probability distributions of various observables, given differential equations, which may be integrated, show how these observables change in time. Keep in mind, we've evolved from thinking the Sun was static, (It is a linear combination of a stationary state[not static], its' probablities don't change!!, and that it revolved around the Eartha linear combination of a stationary state, with no changing probabilities), it's now the other way around.

. For any questions, suggestion or comment, feel free to contact me on GitHub issues or email me at stowino21@gmail.com.

2.1.0.0.1 Axioms of Quantum Mechanics

1. **A1**: The wavefunction is the complete description of a quantum system.
2. **A2**: Hernitian operators are observables.
3. **A3**: The measurement axiom.
4. **A4**: Time evolution via the Schrödinger equation.

2.1.0.0.2 Illustrating Vectors

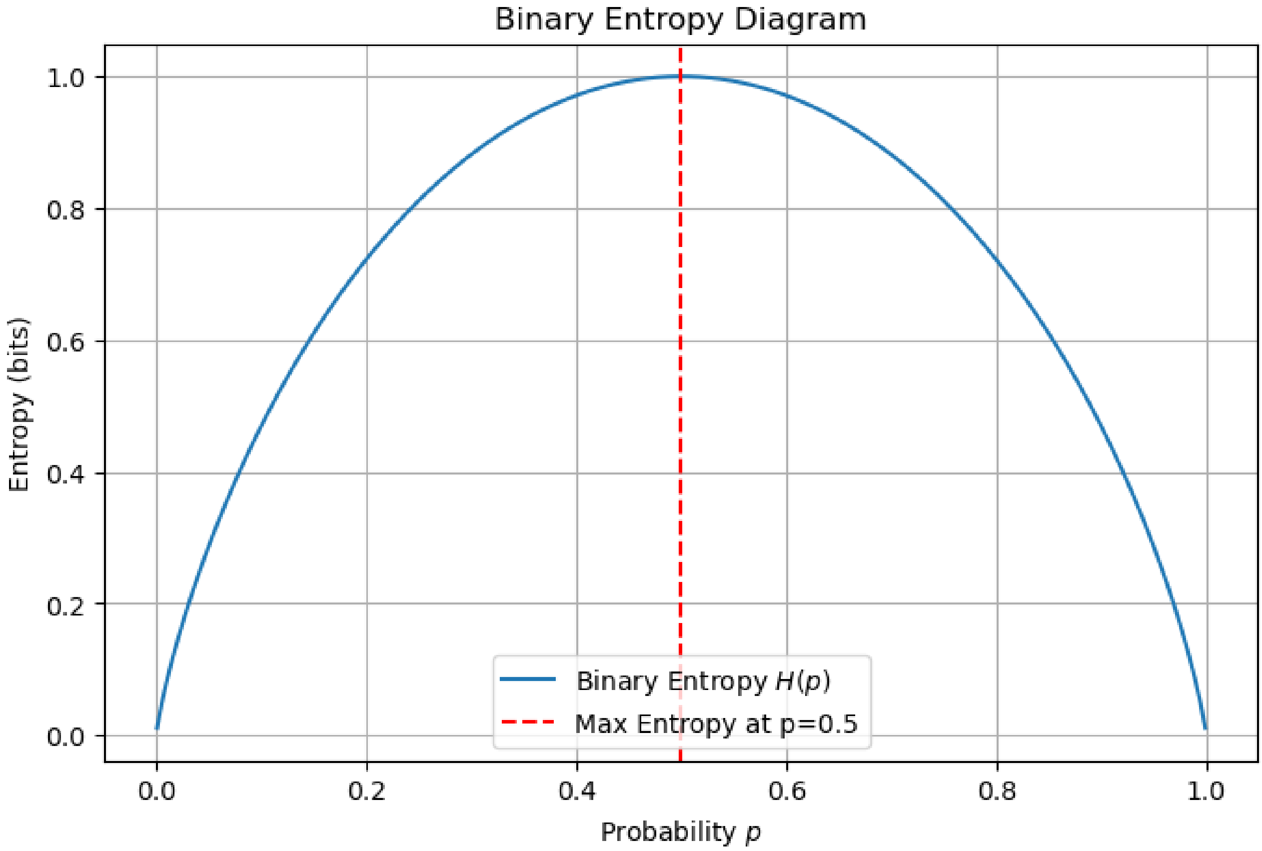

Figure 2.1: Binary Entropy

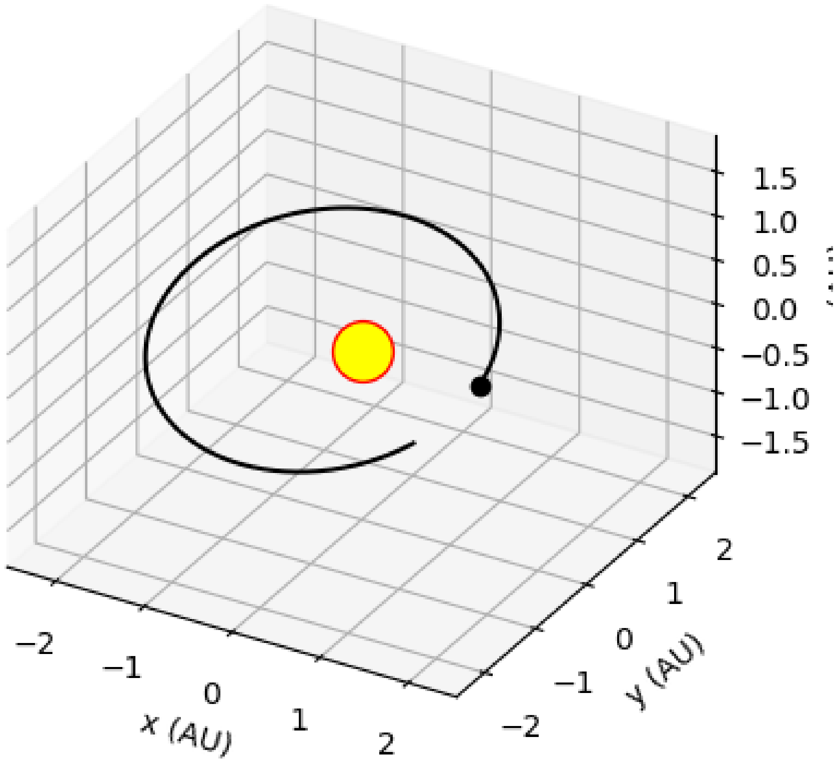

Figure 2.2: Vector pointing towards the Sun

2.2 Some Conceptual Frameworks; `The names in color are Nobel prize receipients, and nomenees.`

- Copenhagen interpretation(s) - Bohr, Heisenberg, Pauli, von Weisäcker, Peierls.
- Non-local Hidden Variables - de Broglie, Bohm, Maudlin, Goldstein, Hardy.
- Convival Solipism - Zwirn
- Modal Interpretations - Kochen, Dieks, van Fraassen, Bub, Sipe.
- Conscienceness-induced Collapse - Wigner, von Neumann, Josephson.
- Objective Collapse Models - Ghiradi-Rimini-Weber, Pearl, Penrose, Tumulka.
- Consistent/Decoherent Histories - Griffiths, Omnés, Gell-Mann, Hartle.
- Superdeterminsm - t'Hooft, Hossenfelder.
- Rational Interpretations - Rovelli
- Ensemble Interpretations - Ballentine, Einstein

• Informational Interpretation - Zellinger, Brukner, Bulb.

• Many-Worlds Interpretation - Everett, Hawking, Science-fiction fans.

Remark

Required reading:

• Spectrum of the harmonic oscillator

• Spectral Theorem

• Linear Algebra(particularly Matrix Theory)

• Continuous Functional Calculus(systems of 2nd order differential equations for vector-valued functions, and factoring)

• Measure Theory

• Equi-partition theorem(law of equi-partition of Energy)

• Electromagnetic theory of light.

Chapter 3 Terminology

3.1 Definitions

Quantum-Theory $\rightarrow$ A generalization/systematic refinement of classical mechanics

Quantum-communication/computation $\longrightarrow$ Manipulating qubits with quantum gates; Qubits being transmitted between different parties; Exploring applications based on sending information, rather than processing it.

Quantum Fourier Transform $\longrightarrow$

$$\phi = \sum_{k=0}^{n-1} y_k \mid k >= \frac{1}{\sqrt{2}} \sum_{k=0}^{n-1} e^{\frac{2\pi ijk}{N}} \mid k >= \frac{1}{\sqrt{2}} \sum_{k=0}^{N-1} \omega^{jk} \mid k > \quad \text{For N = 2,} \quad \text{k = 0,1} \quad \text{and} \quad \omega = e^{\frac{2\pi}{N}}) \tag{3.1}$$

An-entangled-pair $\longrightarrow$ Applying one qubit by manipulating the larger set of the four possible states.

Von Neumann entropy $\longrightarrow$

$$S(\rho) = -tr(\rho \ln \rho) = -\sum_j \lambda_j \ln \lambda_j \text{(These are diagonalized values);} S(\rho) = 0 \iff \rho^2 = \rho \text{(case of a pure state/projector)} \tag{3.2}$$

Maximally mixed state $\longrightarrow$ $S(\rho_m)$ = - N $\frac{1}{N}$ ln $\frac{1}{N}$ = $\ln N$

Gibbs entropy(classical statistical mechanics) $\longrightarrow$

$$S_G = -k_B \sum_i P_i \ln P_i, where \quad k_B \text{is the Bolzmann constant, and P is probabilities} \tag{3.3}$$

Shannon entropy(classical information theory) $\longrightarrow$

$$H(X) = \mathbf{E}(\log_b P(X)) = -\sum_x P(x) \log_b P(x), \text{where H is for Shannon,}\mathbf{E}\text{is the expectation/average value, X is a variable, and P is p} \tag{3.4}$$

State-reduction $\longrightarrow$ The effective two-level Hamiltonian($\hat{H}_{eff}$ = $\begin{pmatrix} 0 & \Omega \\ \Omega & \triangle - i\frac{\Gamma}{2} \end{pmatrix}$), dissipating via spontaneous emission, during "measurement", included in the dynamics(A Quantum trajectory is NOT reversible). Note that $[\hat{H}_{eff}, \hat{H}^{\dagger}_{eff}]$ $\neq$ 0, this Hamiltonian also generates a non-unitary dynamics $e^{-i\hbar_{eff}(t)} \neq e^{i\hat{H}^{\dagger}_{eff}(t)}$

Measurement-of-a-state(indicates a loss of energy) $\longrightarrow$

$$\langle \psi \mid \psi \rangle = \langle \psi_0 \mid e^{-i\hat{H}_{eff}(t)} e^{i\hat{H}_{eff}(t)} \mid \psi_0 \rangle \rightarrow e^{-\Gamma(t)} \tag{3.5}$$

$\log_b$ can take any of the values 2[Classical bits], e[the Natural Logarithm], and Base 10[orders of magnitude]

$H_{bin}(p)$ = $-p \log p - (1-p) \log(1-p) \rightarrow$ e.g. $H_{bin}(\frac{1}{2})$ = $-\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2}$ = $\log 2$

Normalization - technical manipulation of Mathematical symbols(to eliminate the physical vacuum contribution, and to simplify the one electron contribution).

Vacuum-region $\rightarrow$ discontinuity of fields in the small.

See figure 3.2 Limit - $\forall$ total infinities, the degrees of freedom ceases to increase without limit as shorter and shorter time intervals are considered.

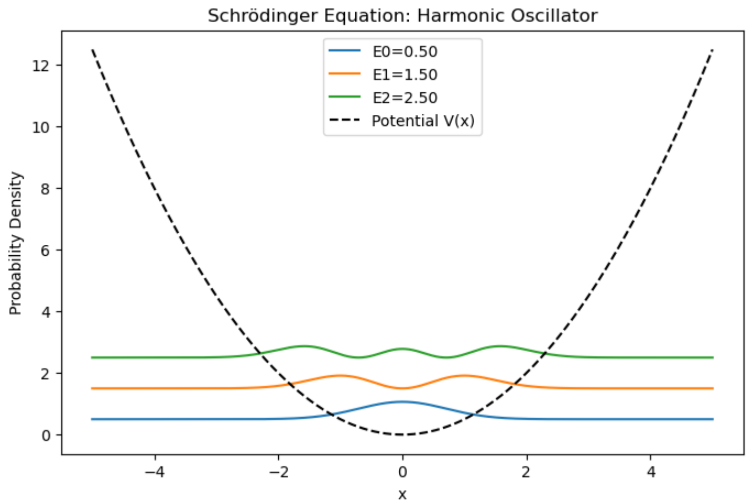

Figure 3.1: This is a figure of the Quantum Harmonic Oscillator. It is defined by $E_n = (n + \frac{1}{2})\hbar\omega$

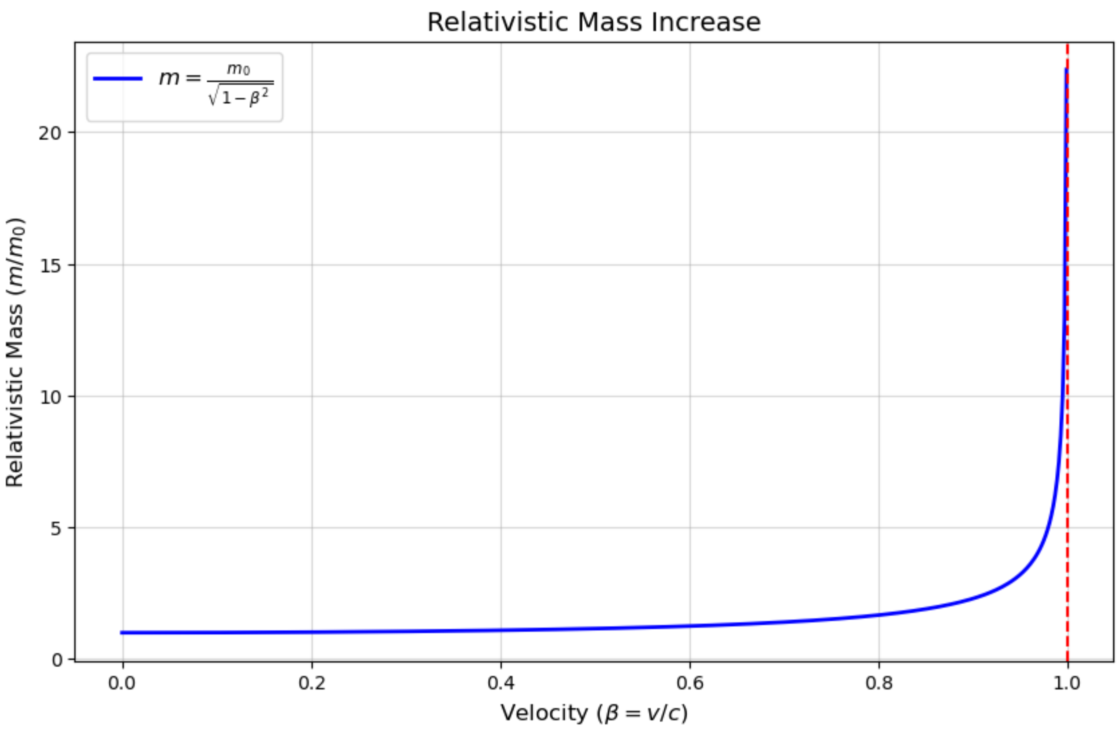

Figure 3.2: Limit concept of Mass with the Speed of Light

Wave-equation $\longrightarrow$

$$\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2} \text{[Relates acceleration and curvature].} \tag{3.6}$$

Decoherence - The inevitable interaction of a system and its environment(quantum-to-classical transition; Its a fundamental limiting factor in the useful operation of Quantum Computers).

Spin(total-angular-momentum)/directional quantization) - a particle ℓ of non-negative integer or half integer. The Hilbert space for such a particle is $\mathscr{L}^2(\mathbb{R}^3)\hat{\otimes}V_\ell$, where V_ℓ is an irreducible projective representation of SO(3) of dimension $2\ell + 1$

Photons - A collection of an indefinite number of indistingishable particles, that obey Bose-Einstein statistics, with integer spin $\not{k}$.

Boson(Bose-Einstein statistics) - A particle with an integer spin

Fermion(Fermi-Dirac statistics) - A particle with half-integer spin e.g. an electron

For j,k,l $\in$ 1,2,3 we define $\epsilon_{j,k,l}$ by the formula:

$$\epsilon_{j,k,l} = \begin{cases} 1 \text{ if j,k,l is an even permutation of } (1,2,3) \\ -1 \text{ if j,k,l is an odd permutation of } (1,2,3) \\ 0 \text{ if any two of j,k,l are equal} \end{cases} \tag{3.7}$$

e.g $\epsilon_{3,2,1}$ = -1 and $\epsilon_{2,1,2}$ = 0.

The commutation relations for the basis F_1, F_2, F_3 for SO(3) may be written(using the summation convention!) as:

$[F_j, F_k] = \epsilon_{j,k,l}F_l$

i.e. if we take j=1 and k=2, then the sum on ℓ gives a nonzero value only when ℓ = 3, and we recover the relation $[F_1, F_2] = F_3$

Invariant $\rightarrow$ The density ν is irreducible under translations in the real directions.

Polarization - The set of directions in which the elements of the quantum subspace are covariantly constant.

Covariant-derivative -$\longrightarrow$ A conection ∇ operator for a vector field.

Integrability $\rightarrow$ Two complex vector fields X and Y lie in P_z at each point z, then so does [X,Y]

Hermitian-line-bundle $\longrightarrow$ (L,∇) connection of $L^{\otimes k}$ having a curvature $\frac{\omega}{\hbar}$ for any positive integer n(things behave nicely when k tends to ∞)

Half-form-space $\longrightarrow$

$$s = (\psi \circ q) \otimes \sqrt{q^*(\beta)} \tag{3.8}$$

A measurable-function ψ: X $\longrightarrow \mathbb{C}$ is said to be integrable, if $\int_X \mid \psi \mid$ d$\mu < \infty$

Density of

$$\mu \quad w.r.t. \quad \nu \longrightarrow \mu(E) = \int_E \rho \quad d\mu \quad \forall E \in \omega \tag{3.9}$$

supposing μ and ν are two σ-finite measures on a measure space(X,Ω), and that μ is absolutely continuos w.r.t. ν: then there exists a non-negative, measurable function ρ on X.

Monotone-class $\longrightarrow$ Suppose $\mathscr{M}$ is a monotone class of subsets of a set X, and suppose $\mathscr{M}$ contains an algebra $\mathscr{A}$ of subsets X, then $\mathscr{M}$ contain the σ-algebra generated by $\mathscr{A}$

Dirac δ function $\longrightarrow$

$$\delta_\chi = \chi(0), roughly \quad \delta(x-X) = \begin{cases} \infty \text{ if x = X} \\ 0 \text{ otherwise} \end{cases} \tag{3.10}$$

The function illustrates the limit of a distribution.

If $\|\cdot\|$: V $\longrightarrow \mathbb{R}$, then

$$\|\psi\| = \sqrt{\langle\psi,\psi\rangle} \tag{3.11}$$

$\|\cdot\|$ is a norm on V

Hilbert-space $\longrightarrow$ A vector space $\mathscr{H}$ over $\mathbb{R}$ or $\mathbb{C}$, equipped with an inner product $\langle\cdot,\cdot\rangle$, such that $\mathscr{H}$ is complete in the norm above. [For a seperable infinite dimensional $\mathscr{H}$, the partially ordered set of all questions in QM(Quantum Mechanics), is isomorphic to the partially ordered set of all closed sub-spaces].

Orthogonal-space $\longrightarrow$ If V is any closed subspace of $\mathscr{H}$, a subspace $V^\perp$ of $\mathscr{H}$, is:

$$V^\perp = \phi \in \mathscr{H} \mid \langle\phi,\psi\rangle = 0 \quad \forall\psi \in V \tag{3.12}$$

Skew-self-adjoint $\longrightarrow$

$$A^* = -A \quad \text{One-parameter group of orthogonal/unitary transformation.} \tag{3.13}$$

Orthogonal-projection $\longrightarrow$ For a closed subspace V, where V = range(P), P is the projection if it is any bounded operator on $\mathscr{H}$ satisfying P^2 = P, and P^* = P

A definite-state(random sequence) $\longrightarrow$

$$\|\bar{\sigma}_z^N \mid \psi\|^2 = \frac{1}{N^2}\langle\psi \mid \sum_{r=1}^{N}\sum_{s=1}^{N}\sigma_z^{(r)}\sigma_z^{(s)} \mid \psi\rangle \tag{3.14}$$

$$\langle\psi \mid \sigma_z^{(r)}\sigma_z^{(s)} \mid \psi\rangle = \delta^{rs} \implies \lim_{N\to\infty}\|\bar{\sigma_z}^N\|^2 = \lim_{N\to\infty}\frac{1}{N^2}N = 0 \tag{3.15}$$

Likewise for $\bar{\sigma}_z^a$ etc.

Zorns'-Lemma $\longrightarrow$

Let P be a set of elements a,b...Suppose there is a binary relation defined between certain pairs (a,b) of elements P, expressed by a $\prec$ b, with the properties: $\begin{cases} a \prec a \\ \text{if a} \prec \text{b and b} \prec \text{a, then a = b,} \\ \text{if a} \prec \text{b and b} \prec \text{c, then a} \prec \text{c(transitivity)} \end{cases}$ Then P is said to be partially ordered or semi-orde

Curl $\longrightarrow$ Measure of the rotation in the vector field about the points in the direction of the normal vector **N**, and Stokes theorem justifies this interpretation.

Amperes Law(line integral along a closed path) $\longrightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \mathbf{l}_{encl}$, where μ_0 is the magnetic constant, and l_{encl} is the net current enclosed by the path.

Equi-partition-principle $\longrightarrow$

$$E = \frac{3}{2}kT, \text{where k is the Boltzmans' constant.} \tag{3.16}$$

See page 43 for Ground-state $\longrightarrow$ An electron in a stable innermost orbit/lowest energy state(most negative), and cannot fall into the nucleus.

Equilibrium position $\longrightarrow$ An electron in a stable innermost orbit/lowest energy state(most negative), and cannot fall into the nucleus. (Its positive definite at its local minimum, hence its Energy is $\frac{\hbar\omega}{2\pi}$ at 0). The equality $\frac{\hbar\omega}{2\pi}$ is the Minimum Energy. ψ_0(x) = $(2\pi\sigma^2)^{-\frac{1}{4}}\ e^{-\frac{x^2}{4\sigma^2}}$, with σ^2 = $\frac{\hbar}{2m\omega}$, and E = $\frac{\hbar\omega}{2}$.

Pauli-exclusion-principle $\longrightarrow$ No two or more electrons in an atom, can have the same value, for any of the four quantum numbers.

Self-adjoint $\longrightarrow T(F(\phi), \psi) = T(F(\psi), \phi) = T(\phi, F(\psi))$

Eigenvalue-equation $\longrightarrow$ An equation, where the operator, operating on a function, produces a constant, times the function. [The function is called an Eigenfunction, and the resulting numeric is called the Eigenvalue]. Eigen here is the German word meaning self, or own.

Hamiltonian-energy(discrete) $\longrightarrow \hat{\mathbf{H}} = \frac{1}{2}m\omega_c^2\hat{r}_c^2$: This is the Kinetic Energy of a particle of Mass m with Cyclotron Frequency ω_c, and radius $\hat{\mathbf{r}_c}$

Dirac equation $\longrightarrow$

$$i\hbar\frac{\partial\psi}{\partial t} = \left[c\alpha \cdot \hat{\mathbf{p}} + \frac{e}{c}\mathbf{A} + \beta mc^2 + V(r)\right]\psi \tag{3.17}$$

Where the coupling of the electron to the scalar potential $\Phi(r)$ is included via V(r) = $-e\Phi(r)$.

Gibbs Inequality $\longrightarrow H(f_1, ..., f_n) \leq -\sum_{j=1}^{m} f_j log(u_j)$

Projection-valued-measureassociated with f(A) $\longrightarrow$ f(A')

[f(A) is the self-adjoint operator, corresponding to the observable $f(A')$].

E $\rightarrow P^A_{f^{-1}(E)}$. Note that the operators corresponding to the Energy and Momentum observables of a quantum system are obtainable as infinitesimal generators of certain one-parameter groups of unitary operators.

Eigenvector of a self-adjoint operator $\longrightarrow$ A state in which the corresponding observable takes on the corresponding Eigenvalue with probability **1**

Von Neumann-density-matrix $\longrightarrow$ The matrix of an operator defining a mixed state.

Differentiable-function ψ [Schrődingers equation] $\longrightarrow$

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2m}{\hbar^2}\left(\mathbf{E} + \frac{e^2}{\sqrt{x^2+y^2+z^2}}\right) = 0, with \quad \hbar = \mathbf{K}. \tag{3.18}$$

Canonical-commutation-relation $\longrightarrow$ [$\hat{X}$, $\hat{P}$] = $i\hbar\delta_{ij}$, i,j = x,y,z.

Heisenberg-commutation-relation $\longrightarrow$

$(X_{\phi_1}, X_{\phi_2} - X_{\phi_2}, X_{\phi_1})$ = $(\frac{\hbar}{i})$ $[f_{\phi_1}, f_{\phi_2}]$, where f_ϕ is an observable, and X_ϕ the self-adjoint operator.

Discrete $\longrightarrow$ A representation which can be decomposed as a direct sum of irreducibles.

Schurs' Lemma(only true in $\mathbb{C}$ representations, NOT $\mathbb{R}$ vector spaces) $\longrightarrow$ Let T be $\in$ R(L,M), then L restricted to the orthogonal complement of the null space of T, is $\equiv$ to M restricted to the Closure the Range of T.

Degeneracy $\longrightarrow$ Occurrence of multiple eigenvalues/states sharing the same energy.

Zeeman-effect $\longrightarrow$ Splitting apart of eigenvalues, hence of spectral lines, in a magnetic field.

Kronocker-delta(Orthonormality condition) $\longrightarrow$ $\delta_{ij} = \begin{cases} 1 \text{ if i = j} \\ 0 \text{ if i} \neq \text{j} \end{cases}$

Tensor/Kronecker-product of two Spin one-half state space[needed to calculate the hyperfine-splitting in the Hydrogen-atom] $\longrightarrow$

$\mid 0, 0\rangle$ = $\frac{1}{\sqrt{2}}$ $(|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 - |\downarrow\rangle_1 \mid \otimes \uparrow_2)$ = $\frac{1}{\sqrt{2}}$ $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$

Clebsch-Gordan-coefficient(expansion)/vector coupling/Wigner $\longrightarrow$

$$\mid J, M\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} \langle JM \mid m_1, m_2\rangle\langle m_1, m_2 \mid J', M'\rangle = \delta_{jj'}\delta_{MM'} \tag{3.19}$$

Hydrogen hyperfine split $\longrightarrow$ Energy shift/difference of the lowest antiparallel state of the spin of the protons magnetic moment and the electrons magnetic field, pointing in the same direction(corresponds to the fine-structure splitting).

Collision-broadening $\longrightarrow$ Distortion of an emitted wave during radiation, thus smearing the spectral line.

See equation (4.4) Inner-product $\longrightarrow$ <A|B> = $\frac{1}{2}tr(A, B^\dagger)$

A Matrix changes a derivative, so A = $fracddt$. To find the transpose of this (A) we need to define the Inner Product between two functions x(t) and y(t).

The Inner Product changes from the sum of X_kY_k to the integral of x(t) y(t).

Inner product of functions X^Ty = (X, y) = $\int_{-\infty}^{\infty}$ x(t)y(t)dt

From this Inner Product, we know the requirement on A^T . The word "adjoint" is more correct than "transpose" when we are working with derivatives.

The transpose of a matrix has $(Ax)^T\mathbf{y}$ = $\mathbf{x}^T$ $(A^T\mathbf{y})$. The adjoint of A = $\frac{d}{dt}$ has:
(Ax,y) = $\int_{-\infty}^{\infty}$ $\frac{d}{dt}$ y(t) dt = (x, A^Ty)
Recognize integration by parts. The derivative moves from the first function x(t), to the second function y(t). During that move, a minus sign appears. This tells us that the transpose of the derivative is minus the derivative.

The derivative is antisymmetric: $\mathbf{A}$ = $\frac{d}{dt}$, and $\mathbf{A^T}$ = $-\frac{d}{dt}$. Symmetric matrices have S^T = S, antisymmetric matrices have A^T = -A. Linear algebra includes derivatives and integrals, because those are both linear.

This anti-symmetry of the derivative applies to centered difference matrices. A =

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

;transposes to A^T =

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

= -A.

And a forward difference matrix transposes to a backward difference matrix, multiplied by -1. In differential equations, the second derivative (Acceleration) is symmetric. The first derivative (Velocity) is antisymmetric.

Entanglement $\longrightarrow$ A Bell state that cannot be expressed as a tensor product of a single qubit state with another single qubit state.

No-cloning-theorem $\longrightarrow$ One cannot create a copy of quantum states, although teleporting is permitted.

Teleportation $\longrightarrow$ Same quantum state in different locations.

See 3.1 for the Quantum-Harmonic-Oscillator $\longrightarrow$
$\hat{H} = \hbar\omega\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right) = \hbar\omega\left(\hat{N} + \frac{1}{2}\right)$, where $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$;
$\hat{a}^{\dagger} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$, and
$\hat{N} = \hat{a}^{\dagger}\hat{a} = \frac{m\omega}{2\hbar}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$

Flux-quantum(a dimensionless quantity) $\longrightarrow \Phi = \frac{\hbar}{2e}$, where 2e is the charge of the electron(Cooper pairs).

Von Neumann-equation $\longrightarrow \partial_t\ \rho(t) = -\frac{i}{\hbar}\ [H, \rho(t)]$, where ρ is the density matrix, and H is the Hamiltonian; similar to the Schr$\ddot{o}dinger$ equation, $i\hbar\partial_t\rho(t) = [H, \rho(t)]$(describes the time-evolution of the density operator).

Resolution-of-an-dentity $\longrightarrow \sum_{k_0}\sum_{\mathbf{k}_1} < k_0|A_0|\mathbf{k}_1 >< \mathbf{k}_1|A_1|k_0 >$

3.2 Ideas

Ideas that work[linear, and angular momentum of a field

• A perfect crystal lattice is effectively empty, for an electron in the lowest band, even though the space is full of atoms

• The absolute value of Energy is equivalent to the fraction of a reduced mass of a proton-electron system, times Charge to the fourth power, over twice an integer squared times the square of the Plancks' constant $\longrightarrow |E| = \frac{\mu Q^4}{2n^2\hbar^2}$

• There are two special classes of polarizations(See 9), those that are purely real(i.e.,$\bar{P}_z = P_z\ \forall$ z $\in$ N), and those that are purely complex(i.e., $P_z \cap \bar{P} = 0\ \forall$ z $\in$ N).

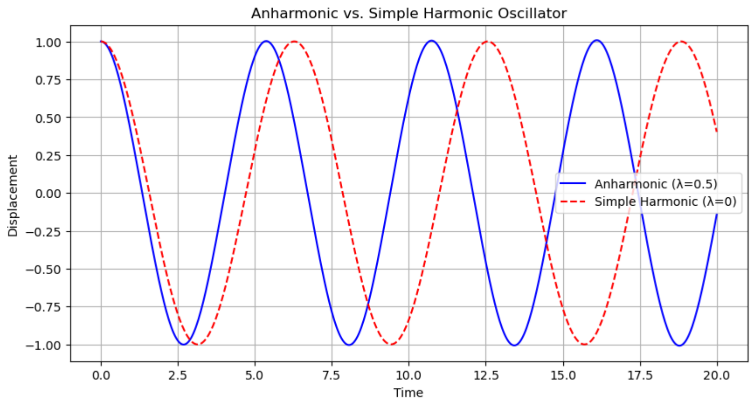

Figure 3.3: Representation of Anharmonic vs simple Harmonic Oscillator

• The half-form of the Hilbert space is the completion w.r.t. $|s|^2 = \int_\Xi \widetilde{(s1, s2)}$ of the space of polarized for which $|s|^2 < \infty$

• Pairing maps $\longrightarrow \int_{\mathbb{R}} \overline{\int_{\mathbb{R}} \phi(x) e^{-\frac{ixp}{\hbar}} dx}\ \psi$ (p) dp

• Dirac δ function $\longrightarrow \delta_x i = \xi(0)$

• Faradays Law(The Electric field is the negative of the rate of change of the corresponding Magnetic field w.r.t time) $\longrightarrow curl\mathbf{E}$ = - $\frac{\partial \mathbf{B}}{\partial t}$

Linear-system $\longrightarrow$ Its properties satisfy a finite-dimensional vector space, has a posive-definite quadratic form, and is independent.

Configuration-vector $\phi \longrightarrow \frac{d}{dt}(\psi, \phi) = \psi, -I^*(\phi)$ or $\frac{d\phi}{dt} = \psi\ \frac{d\psi}{dt} = -I^*(\phi)$, satisfies the second-order equation $\frac{d^2\phi}{dt}$ $= -I^*(\phi)$, where ϕ is a scalar or vector-valued function on 3-space, and I^* is a differential operator.

• **Physical-law** (Seiberg-Witten-equations(useful on sympletic 4-manifolds)), is an example of a class of ϕs"(a scaling-factor). proves one suitable existance and uniqueness theorems of the class of ϕs') $\longrightarrow$ When ϕ is regarded as a function of x,y,z and t, then it becomes a partial differential equation of the form $\frac{\partial^2\phi}{dt^2} = -I^*(\phi)$.

• Light $\longrightarrow$ Vibrations in the electromagnetic-field

Energy $\longrightarrow$ An observable corresponding to $\hbar$, times the dynamical operator, and that its an integral so that we have an **energy conservation law**.

• Heisenberg-commutation-relation $\longrightarrow Q_j Q_k - P_j P_k = i\ \hbar\ \delta_j^k$

• Wavefunction of a system $\longrightarrow$ A state vector ψ, is a square summable function on $\mathscr{M}$, and trajectory $e^{-iHt}(\psi)$ is a function on $\mathscr{M} \times \mathbf{R}$, where $\mathbf{R}$ is the real line(This is a function of $\mathbf{3n+1}$ real variable).

• Expected-value of the j-th Momentum component $\longrightarrow \frac{\hbar}{i} \int \left(\frac{\partial\psi}{\partial q_j}\overline{\psi}\right) dq_1 ... dq_{3n}$.

Chapter 4 Physics of the Universe

4.1 Historical Timeline of the Quantization of Light

Table 4.1: Historical Timeline of the Quantization

Year	Description
1670–1678	**Isaac Newton** proposes the corpuscular theory of light, suggesting light consists of particles.
1801	**Thomas Young** performs the double-slit experiment, demonstrating the wave nature of light.
1864	**James Clerk Maxwell** formulates the theory of electromagnetism, describing light as an electromagnetic wave.
1900	**Max Planck** introduces the concept of quantization to explain blackbody radiation, proposing that energy is emitted or absorbed in discrete packets called *quanta*.
1905	**Albert Einstein** explains the photoelectric effect using Plancks' quantum hypothesis, suggesting that light consists of discrete packets of energy called *photons*

where:

E = h ν

4.1.1 Key Takeaways

The quantization of light marked a paradigm shift in Physics, leading to the development of Quantum Mechanics. Light exhibits both wave-like and particle-like properties, a concept known as **wave-particle duality**.

4.2 Manifolds of pairing maps on an invariant self-adjoint subspace

4.2.1 Constants

A few constants:

$\hbar$*c $\simeq$197.327 MeV $\cdot$ fm

Coupling/Fine structure constant $\alpha \to \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137.036}$, $m_e\ c^2 \simeq$ 0.511 MeV, $m_p\ c^2 \simeq$ 9.38 MeV. Described as deviations in the electron spin/removal of energy degeneracy of the orbital quantum number ℓ: E = $E_{n,\ell}$[Energy shifts, and splitting of spectral lines].

Bohr Radius $\longrightarrow r_B = \frac{4\pi\epsilon_0\hbar^2}{e^2 m_e} \approx 0.53 \times 10^{-10}$m

4.2.2 Useful Formulae

- (1) Relativity: p = m$v\gamma$, E = γ m c^2, $E^2 = p^2 * c^2 + m^2\ c^4$, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$, $\beta = \frac{v}{c}$

- (2) de Broglie: $\lambda = \frac{h}{p}$,

Compton: $\lambda_C = \frac{2\pi h}{mc}(1-\cos\theta)$,

$\hat{p} = \frac{h}{i}\ \frac{\partial}{\partial x}$,
$\hat{\mathbf{p}} = \frac{\hbar}{i}\ \nabla$,
$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$,
$[\hat{p}_i, f(\hat{\mathbf{x}})] = \frac{\hbar}{i}\ \frac{\partial f}{\partial x_i}$.

The equations above illustrate Photo-Electron Energy and interaction free measurements.

- (3) Schr$\ddot{o}$dinger-equation:

i $\hbar\ \frac{\partial\psi}{\partial(t)}$ (x,t) = ($\frac{-i\hbar^2}{2m}\ \nabla^2$ + V (x,t)) $\psi(x,t)$,
$\frac{\partial}{\partial(t)}\ \rho(x,t) + \nabla\cdot\mathbf{J}$ (x,t) = 0,
ρ (x,t) = | $\Psi(x,t)|^2$,
$\mathbf{J}$ (x,t) = $\frac{\hbar}{m}$ Im $[\psi^*\ \nabla\psi]$.

- (4) Fourier transforms: $\psi x = \frac{1}{\sqrt{2\pi}}\int dk\ \phi$ (k) e^{ikx},

$\Phi(k) = \frac{1}{\sqrt{2\pi}}\int dx\ \Psi$ (x) e^{-ikx}, $\int$dx| $\Psi(x)|^2 = \int dk$ | $\Phi k|^2$. $\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ikx}dx = \delta(k)$, $\frac{1}{(2\pi)^3}\int e^{ik}\cdot x(d)^3x = \delta^{(3)}(k)$.

From the distribution-integral $\rightarrow$

$$\int_{-\infty}^{\infty} e^{(-ax^2+bx)}dx = \sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a}}, \quad when \quad \Re(a) > 0. \tag{4.1}$$

- (5) expectation value:

$\langle\hat{\mathbf{Q}}\rangle \equiv \langle\psi,$
$\hat{\mathbf{Q}}\psi\rangle$, $i\hbar\frac{d}{dt}\langle\hat{\mathbf{Q}}\rangle = \langle[\hat{\mathbf{Q}}, \mathbf{H}]\rangle$, $\hat{\mathbf{Q}}$ time independent.

- (6) Uncertainty:

Consider Schwarz Inequality. Since $[\sigma_Q^2, \sigma_R^2] \geq (\frac{1}{2i}\langle[\hat{Q}, \hat{R}])^2$,

$$[\sigma_x^2, \sigma_p^2] = (\frac{1}{2i}\langle i\hbar\rangle)^2 \implies (\frac{[i\hbar]}{2i})^2 \tag{4.2}$$

Resulting to $[\frac{\hbar^2}{4}]$

So, $\boxed{[\sigma_x^2, \sigma_p^2] = \frac{\hbar}{2}}$

$\triangle\ A \equiv |A - \langle\hat{A}\rangle\mathbf{1}\psi|$

$(\triangle\ A)^2 = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2 \geq 0.$

$\triangle\ A\ \triangle\ B \geq \langle\psi \quad |\ \frac{1}{2i}\ [\hat{A}, \hat{B}] \quad |\ \psi\ \rangle$

- (7) saturation: $(\hat{B} - \langle\hat{B}\rangle\not{\Bbbk})\ |\ \psi\rangle$ = i $\lambda(\hat{A} - \langle\hat{A}\not{\Bbbk})\ |\ \psi\rangle$, $\lambda \in \Re$.

$\triangle\ x\ \triangle\ p \geq \frac{\hbar}{2}$. For $\psi\ \ \exp(-\frac{1}{4}\frac{x^2}{\triangle^2})$,
$\triangle\ x = \triangle$ and $\triangle\ p = \frac{\hbar}{2\triangle}$.

$\triangle\ H\ \triangle\ t \geq \frac{\hbar}{2}$, $\triangle\ t \equiv \frac{\Delta Q}{|\frac{dQ}{dt}|}$.

- (8) stationary state: ψ(x,t) = $\psi(x)e^{-\frac{iEt}{\hbar}}$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x) + \mathbf{V}(\mathbf{x})\Psi(\mathbf{x}) = E\psi(\mathbf{x}) \tag{4.3}$$

- (9) variational-principle: $E_{gs} \leq \langle\psi, \hat{H}\psi\rangle$, for all normalized ψ.

- (10) Hellmann-Feynmann: $\hat{H}\ (\lambda)\ \psi_n\ (\lambda)$ = $E_n\ (\lambda)\ \psi_n\ (\lambda) \longrightarrow \frac{dE_n(\lambda)}{d\lambda}$ = $\langle\psi_n(\lambda), \frac{d\hat{\mathbf{H}}\lambda}{d\lambda}\Psi_n(\lambda)\rangle$.

- (11) virial theorem: one dimension: $\langle\ \frac{\hat{p}^2}{2m}\ \rangle$ = $\frac{1}{2}\ \langle\hat{x}\ \frac{dV}{dx}\ \rangle$

- (12) central potential: $\langle\frac{\hat{p}^2}{2m}\rangle$ = $\frac{1}{2}\ \langle r\ \frac{\partial V}{\partial r}\rangle$.

- (13) commutator identities:

[A,BC] = [A,B]C + B[A,C]

$$e^A B e^{-A} = e^{ad_A}B = B + [A,B] + \frac{1}{2!}[A,[A,B]] + \frac{1}{3!}[A,[A,[A,B]]] + ..., \tag{4.4}$$

This is Hadamards' Lemma using Taylors' expansion, with

$$(ad_A)\text{as an adjoint action;} \quad (ad_A(X) = [A,X]) \tag{4.5}$$

Note that $e^A\ e^B \neq e^{A+B}$ = e^C, with C = A + B + $\frac{1}{2}$[A,B] + $\frac{1}{12}$([A,[A,B]] - $\frac{1}{12}$[B[A,B]]) + ..., Its' not evident what the higher terms(...) refer to, although its' been expanded and expressed in terms of commutators.

$e^A B e^{-A}$ = B + [A,B], if [A,[A,B]] = 0,

$[B, e^A]$ = $[B,A]e^A$, if [[A,[A,B]] = 0,

e^{A+B} = $e^A e^B e^{-\frac{1}{2}|A,B|}$ = $e^B e^A e^{\frac{1}{2}[A,B]}$ if [[A,B],A] = [[A,B],B] = 0.

- (14) matrix exponential: $e^{iM\theta}$ = $\mathbf{1}\cos\theta$ + i M $\sin\theta$,

- (15) index-manipulation:

$\mathbf{a}\cdot\mathbf{b}$ = $a_i b_i$,
$\delta_{ij}B_j$ = B_i, δ_{ii} = 3,
$(\mathbf{a}\times\mathbf{b})_i$ = $\epsilon_{ijk}\mathbf{a}_j\mathbf{b}_k$, ϵ 123 = 1
$\epsilon_{ijk}\epsilon_{ipq}$ = $\delta_{jp}\delta_{kq}$ - $\delta_{jp}\delta_{kp}$,

$$\epsilon_{ijk}\epsilon_{ijq} = 2\delta_{kq} \tag{4.6}$$

- (16) linear algebra: matrix representation in the basis $(v_{1,...,}v_n)$: Tv_i = $\sum_k T_{kj}\ v_k$.

- (17) basis change:

u_k = $\sum_j A_{jk}v_j$,

$$T(u) = A^{-1}T(v)A \tag{4.7}$$

• (18) Inner-product:

$$\langle v, v\rangle \geq 0, forall v, \langle v, v\rangle = 0 \iff v = 0, \langle u, v\rangle = \langle vu\rangle^* \tag{4.8}$$

• (19) Schwarz inequality:

$$\mid (u, v) \mid \leq \|u\| \quad \|v\| \tag{4.9}$$

• (20) Complex-vector-space:

$\langle v, Tv\rangle$ = 0,

$\forall\, \nu \in$ V $\longrightarrow$ T = 0.

• (21) adjoint:

$\langle u, Tv\rangle$ = $\langle T^\dagger u, v\rangle$,

$(T^\dagger)^\dagger$ = T,

$(ST)^\dagger$ = $T^\dagger S^\dagger$.

• (22) Hermitian operator T: $T^\dagger$ = T

• (23) unitary operator U: $U^\dagger U$ = $UU^\dagger$ = $\mathbf{1}$.

• (24) orthogonal-projector: P_u: V $\longrightarrow$ U,

$P_u P_u$ = P_u,

$P_u^\dagger$ = P_u,

V = range $P_u \oplus$ null P_u.

• (25) bra-kets:
$\langle\ u \mid v\rangle \equiv \langle u, v\rangle$,

$\mid Tv\rangle \equiv T \mid v\rangle$,

$\langle Tv \mid$ = $\langle v \mid T^\dagger$,

$\langle u \mid T^\dagger v\rangle$ = $\langle Tu \mid v\rangle$.

$\mid \alpha_1 v_1$ + $\alpha_2 v_2\rangle$ = $\alpha_1 \mid v_1\rangle$ + $\alpha_2 \mid v_2\rangle$,

$\langle \alpha_1 v_1$ + $\alpha_2 v_2 \mid$ = $\alpha_1^* \langle v_1 \mid$ + $\alpha_2^* \langle v_2 \mid$.

T = $| u\rangle \langle w | \longrightarrow T^{\dagger}$ = $| w\rangle \langle u |$,

trT = $\langle \mathbf{w} \mid \mathbf{u}\rangle$.

• (26) orthonormal-basis $| i\rangle$:

$\langle i \mid j\rangle = \delta_{ij}$,

$\mathbf{1} = \sum_i \mid i\rangle\langle i \mid$.

$T_{ij} = \langle i \mid T \mid j\rangle \longleftrightarrow$ T = $\sum_{i,j} T_{i,j} \mid i\rangle \langle j \mid$,

$\langle i \mid T^{\dagger} \mid j\rangle^{*}$,

$(T^{\dagger})_{ij} = (T_{ji})^{*}$.

$[M, M^{\dagger}] = 0 \longleftrightarrow$ M is unitarily diagonizable.

• (27) Spectral-Theorem:

$T^{\dagger}$ = T $\longrightarrow$ T = $\sum_k \lambda_k P_k$,

$P_k^{\dagger} = P_k$,

$P_k P_l = \delta_{kl} P_l$,

$\sum_k P_k = \mathbf{1}$.

• (28) Schr$\ddot{o}$dinger picture:

$| \psi, t\rangle = U(t, t_0)|\psi, t_0\rangle$,

$\hat{H}(t) = i\hbar \frac{\partial \mathcal{U}(t,t_0)}{\partial t} \mathcal{U}(t_0, t)$,

U(t,0) = exp$^{(-\frac{i\hat{\mathbf{H}}t}{\hbar})}$, for $\hat{H}$ time independent.

• (29) Heisenberg picture:

$\hat{A}_H(t) \equiv U^{\dagger}(t, 0)\hat{A}_S U(t, 0)$,

$i\hbar \frac{d\hat{A}_H(t)}{dt} = [\hat{A_{\mathbf{H}}}(t)]$,

$\hat{H}_H(t)] + i\hbar\ \frac{\partial \hat{A}_H(t)}{dt}$.

• (30) harmonic oscillator:

$\hat{H} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2 = \hbar\omega(\hat{N} + \frac{1}{2})$,

$\hat{N} = \hat{a}^\dagger \hat{a}$,

$\hat{a} = \frac{1}{\sqrt{2}L_0} (\hat{x} + \frac{i\hat{p}}{m\omega})$,

$\hat{a}^\dagger = \frac{1}{\sqrt{2}L_0}(\hat{x} - \frac{i\hat{p}}{m\omega})$, $L_0^2 = \frac{\hbar}{m\omega}$,

$\hat{x} = \frac{L_0}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger)$,

$\hat{p} = \frac{i}{\sqrt{2}} \frac{\hbar}{L_0}(\hat{a}^\dagger - \hat{a})$,

$[\hat{a}, \hat{a}^\dagger] = 1$,

$[\hat{N}, \hat{a}] = -\hat{}a$,

$\hat{a}\varphi_0 = 0$,

$\varphi_0(x) = N_0 \exp(-\frac{1}{2}\frac{x^2}{L_0^2})$,

$N_0^2 = \frac{1}{\sqrt{\pi}L_0}$,

$\varphi_n = \frac{1}{\sqrt{n!}}(a^\dagger)^n \varphi 0$,

$\hat{H}\varphi_n = \hbar\omega(n + \frac{1}{2})\varphi_n$,

$\hat{\mathbf{N}}\varphi_n = n\varphi_n$,

$\langle \varphi m, \varphi n\rangle = \delta_{mn}$,

$\hat{a}^\dagger \varphi_n = \sqrt{n+1}\varphi_n + 1$,

$\hat{a}\varphi_n = \sqrt{n}\varphi_{n-1}$,

$\hat{x}_H(t) = \hat{x}\cos\omega t + \frac{\hat{p}}{m\omega}\sin\omega t$,

$\hat{p}_H(t) = \hat{p}\cos\omega t - m\omega\hat{x}\sin\omega t$.

• (31) coherent-states:

$\mid \alpha\rangle \equiv e^{\alpha a^\dagger - \alpha^* a} \mid 0\rangle$,

$$\alpha = \frac{\langle \hat{x}\rangle}{\sqrt{2}L_0} + i\frac{\langle \hat{p}\rangle L_0}{\sqrt{2}\hbar} \in \mathbb{C} \tag{4.10}$$

$\hat{a} \mid \alpha\rangle = \alpha \mid \alpha\rangle$,

$\mid \alpha\rangle = e^{-\frac{1}{2}|\alpha|^2}\, e^{\alpha}\, a^\dagger \mid 0\rangle$,

$\mid \alpha, t\rangle = e^{-i\omega\frac{t}{2}} \mid e^{-i\omega t}\alpha\rangle$.

• See equation (4.2.2) (32) orbital-angular-momentum

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \tag{4.11}$$

$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z,$

$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x,$

$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y,$

$\hat{L}^2 \equiv \hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z, [\hat{L}^2, \hat{L}_i] = 0.$

$\nabla^2 = \frac{\mathbf{1}}{r} \frac{\partial^2}{\partial \mathbf{r}^2}\ \mathbf{r}$ - $\frac{\mathbf{1}}{\mathbf{r}^2} \frac{\hat{L}^2}{\hbar^2},$

$\hat{L}^2 = -\hbar^2\ (\frac{\partial^2}{\partial\theta^2} + \cot\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2}),$

$\hat{L}_z = \frac{\hbar}{\beth} \frac{\partial}{\partial\theta},$

$\hat{L}_\pm = \hbar e^{\pm i\phi}\ (\pm\frac{\partial}{\partial\theta} + \mathrm{i}\cot\theta \frac{\partial}{\partial\theta}).$

$Y_{\ell m}\ (\theta,\phi) = \langle\theta\phi \mid \ell m\ \rangle = Y_{\ell m}\ (\Omega) \equiv \mathcal{N}_{\ell m}\ P_\ell^m\ (\cos\theta)\ e^{im\theta}.$

$L_z Y_{\ell m} = \hbar\ \mathbf{m}\ \mathbf{Y}_{\ell m},$

$\mathbf{L}^\angle\ Y_{\ell m} = \hbar^\angle\ \ell\ (\ \ell + 1)\ Y_{\ell m},$

$\int d\Omega \mathbf{Y}^*_{\ell' m'}(\Omega) = \delta_{\ell'\ell}\delta_{m'm},$

$\sum_{\ell=0}^\infty \sum_{m=-\ell}^\ell Y^*_{\ell m}(\Omega) = \delta\ (\cos\theta - \cos\theta')\ \delta\ (\phi - \phi').$

$Y_{0,0}(\theta,\phi) = \frac{1}{\sqrt{4\pi}},$

$Y_1, \pm 1(\theta,\phi) = \mp\sqrt{\frac{3}{8\pi}} \sin\theta\ e^{\pm i\phi},$

$Y_{1,0}\ (\theta,\phi) = \sqrt{\frac{3}{4\pi}} \cos\theta.$

• (33) central potentials:

$\psi(r,\theta,\phi) = \frac{u(r)}{r}\ Y_{\ell m}(\theta,\phi),$

$(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \mathrm{V(r)} + \frac{\hbar^2\ell(\ell+1)}{2mr^2})u(r) = \mathrm{u(r)}, \mathrm{u(r)} \approx r^{\ell+1}$, as r $\longrightarrow$ 0.

• (34) Spin one-half:

$\hat{H} = -\mu \cdot B, \hat{\mu} = g\frac{e\hbar}{2mc} \frac{1}{\hbar}\hat{S},$

$\mu_B = \frac{e\hbar}{2m_e C}, \hat{\mu_e} = \text{-}2\ \mu_B\ \frac{\hat{\mathbf{S}}}{\hbar}.$

$| \, 1\rangle \equiv | \, Z;+\rangle = | +\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$,

$| \, 2\rangle \equiv | \, Z;-\rangle = | -\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$,

$\hat{S}_i = \frac{\hbar}{2}\sigma_i$, $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$,

$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k$,

$\sigma_i\sigma_j = \delta_{ij}\mathbf{1} + i\epsilon_{ijk}\sigma_k$, $(\sigma\cdot\mathbf{a})(\sigma\cdot\mathbf{b}) = \mathbf{a}\cdot\mathbf{b}\mathbf{1} + i\sigma\cdot(\mathbf{a}\times\mathbf{b})$,

$e^{i\mathbf{a}\cdot\sigma} = \mathbf{1}\cos\mathbf{a} + i\sigma\cdot\hat{\mathbf{a}}\sin a$,

$\mathbf{a} = \hat{\mathbf{a}}a$, a = $|\mathbf{a}|$.

$\hat{S_n} \equiv \mathbf{n}\cdot\hat{S} = \frac{\hbar}{2}\mathbf{n}\cdot\sigma$, $\hat{S_n} \, | \, \mathbf{n};\pm\rangle = \pm\frac{\hbar}{2} \, | \, \mathbf{n};\pm\rangle$,

$| \, \mathbf{n}\rangle \equiv | \, \mathbf{n};+\rangle$,

$\langle\mathbf{n} \, | \, \hat{\mathbf{S}} \, | \, \mathbf{n}\rangle = \frac{\hbar}{2}\mathbf{n}$,

$| \, \mathbf{n};_\rangle = \cos\frac{\theta}{2} \, | +\rangle + \sin\frac{\theta}{2}\mathbf{e}^{\mathbf{i}\theta} \, | -\rangle$,

$| \, \mathbf{n};-\rangle = -\sin\frac{\theta}{2}\mathbf{e}^{-}\mathbf{i}\theta \, | +\rangle + \cos\frac{\theta}{2} \, | -\rangle$,

$\hat{R_n}(\alpha) \equiv \exp(-\frac{i}{\hbar}\alpha\hat{S_n})$,

$\hat{R_n}(\alpha) \, | \, \mathbf{n}'\rangle = | \, n''\rangle$, with $\mathbf{n}'' = \mathcal{R}_{\mathbf{n}}(\alpha)\mathbf{n}'$

$\hat{R_n}(\alpha)\hat{\mathbf{S}}\hat{R_n}(\alpha) = \mathcal{R}\,(\alpha)\,\hat{\mathbf{S}}$

• (35) Spin precession: $\hat{H} = \omega_L\cdot\hat{S}$,

$\omega_L = -\gamma\mathbf{B}$, $\hat{\mu} = \gamma\hat{\mathbf{S}}$.

• (36) general angular momentum:

$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{\mathbf{J}}_k \longleftrightarrow \hat{\mathbf{J}}\times\hat{\mathbf{J}} = i\hbar\hat{\mathbf{J}}$,

$[\hat{\mathbf{J}}^2, \hat{\mathbf{J}}_i] = 0$.

$\hat{\mathbf{J}_\pm} = \hat{\mathbf{J}}_x \pm i\hat{\mathbf{J}}_y$, $(\hat{J_\pm})^\dagger = \hat{\mathbf{J}_\mp}$,

$\hat{\mathbf{J}}_x = \frac{1}{2}(\hat{\mathbf{J}_+} + \hat{\mathbf{J}_-})$, $\hat{\mathbf{J}}_y = \frac{1}{2i}(\hat{\mathbf{J}_+} - \hat{\mathbf{J}_-})$,

$[\hat{\mathbf{J}}_z, \hat{\mathbf{J}_\pm}] = \pm\hbar\hat{\mathbf{J}_\pm}$, $[\hat{\mathbf{J}_+}, \hat{\mathbf{J}_-}] = 2\hbar\hat{\mathbf{J}}_z$,

$[\hat{\mathbf{J}}^2, \hat{\mathbf{J}_\pm}] = 0$,

$\hat{\mathbf{J}}^2 = \hat{\mathbf{J}_+}\hat{\mathbf{J}_-} + \hat{\mathbf{J}}_z^2$ - $\hbar\,\hat{\mathbf{J}}_z = \hat{\mathbf{J}_-}\hat{\mathbf{J}_+} + \hat{\mathbf{J}}_z^2 + \hbar\,\hat{\mathbf{J}}_z$.

$\hat{\mathbf{J}}^2|\mathbf{jm}\rangle = \hbar^2\mathbf{j}(\mathbf{j}+1)|\mathbf{jm}\rangle = \hbar\mathbf{m}|\mathbf{jm}\rangle$,

$\mathbf{m} = -\mathbf{j}, ..., \mathbf{j}$

$\hat{\mathbf{j}_\pm}|\mathbf{j}m\rangle = \hbar\,\sqrt{\mathbf{j}(\mathbf{j}+1) - \mathbf{m}(\mathbf{m}\pm 1)}\;|\mathbf{j}, \mathbf{m}\pm 1\rangle$.

$\hat{R_n^\dagger}(\alpha)\hat{\mathbf{J}}\hat{R}_n(\alpha) = \mathcal{R}_n(\alpha)\hat{\mathbf{J}}$,

$\hat{\mathbf{R}_n}(\alpha) = e^{-i\frac{\alpha}{\hbar}\mathbf{n}\cdot\mathbf{J}}$.

• (37) addition of angular momentum:

$\hat{\mathbf{J}} = \hat{\mathbf{J}_1} + \hat{\mathbf{J}_2}$.

• (38) uncoupled basis: $|j_1 j_2; \mathbf{m}_1\mathbf{m}_2\rangle$ CSCO: $(\hat{\mathbf{J}}_1^2, \hat{\mathbf{J}}_2^2, \hat{J}_{1z}, \hat{j_{2z}})$

• (39) coupled basis:

$|\,j_1 j_2;\mathbf{jm}\rangle$ CSCO:

$(\hat{\mathbf{J}}_1^2, \hat{J}_2^2, \hat{\mathbf{j}}^2, \hat{j}_z)$.

$j_1 \otimes j_2 = (j_1 + j_2) \oplus (j_1 + j_2 - 1) \oplus ...\oplus |\,j_1 - j_2\rangle$.

$$|\,\mathbf{j}_1\mathbf{j}_2;\mathbf{jm}\rangle = \sum_{\mathbf{m}_1+\mathbf{m}_2=\mathbf{m}} |\,\mathbf{j}_1\mathbf{j}_2;\mathbf{m}_1\mathbf{m}_2\rangle \underbrace{\langle \mathbf{j}_1\mathbf{j}_2;\mathbf{m}_1\mathbf{m}_2\,|\,\mathbf{j}_1\mathbf{j}_2;\mathbf{jm}\rangle}_{\text{Clebsh-Gordan- coefficient}} \tag{4.12}$$

$\hat{\mathbf{J}}_1 \cdot \hat{\mathbf{J}}_2 = \frac{1}{2}\,(\hat{J_{1+}}\hat{J_{2-}} + \hat{J_{1-}}\hat{J_{2+}}) + \hat{J_{1z}}\;\hat{J_{2z}} = \frac{1}{2}\,(\hat{\mathbf{J}}^2 - \hat{\mathbf{J}}_1^2 - \hat{\mathbf{J}}_2^2)$.

• (40) Hydrogen-atom:

$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mathbf{m}}$ - $\frac{Ze^2}{r}$, $Z = 1$ for hydrogen.

$E_n = -\frac{Z^2 e^2}{2a_0}\frac{1}{\mathbf{n}^2}$, $a_0 = \frac{\hbar^2}{me^2} \simeq 52.9 \pm$,

$\frac{e^2}{2a_0} = \frac{1}{2}\mathbf{mc^2}\alpha^2 \equiv$ Ry $\simeq 13.6$ eV.

$\psi_{n\ell m}(\mathbf{X}) = \mathcal{N}(\frac{r}{a_0})^\ell\;(polynomial \in \frac{r}{a_0} of degree[\mathbf{n} - (\ell+1)])\;e^{-\frac{Zr}{\mathbf{na_0}}}\;\mathbf{Y}_{\ell m}\;(\theta, \phi)$,

$\psi_{100}(\mathbf{r}, \theta, \phi) = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-\frac{Z}{a_0}}$

$Z = 1$: $\langle r \rangle = \frac{1}{2} a_0 (3n^2 - \ell(\ell+1))$, $\langle \frac{1}{r} \rangle = \frac{1}{a_0 n^2}$,

$\langle \frac{1}{r^2} \rangle = \frac{1}{a_0^2 n^3 (\ell + \frac{1}{2})}$,

$\langle \frac{1}{r^3} \rangle = \frac{1}{\mathbf{a^3 n^3} \ell(\ell+\frac{1}{2})(\ell+1)}$.

• (41) Fine structure corrections of the Hydrogen atom:

$E_{n\ell\mathbf{j}m_j} = -\frac{e^2}{2\mathbf{a}_0} \frac{1}{\mathbf{n}^2} (1 + \frac{\alpha^2}{\mathbf{n}^2} [\frac{\mathbf{n}}{\mathbf{j}+\frac{1}{2}} - \frac{3}{4}])$.

• (42) Density matrix:

$E = [(\mathbf{p}_1, \mid \psi_1\rangle), ..., (p_n, \mid \psi_n\rangle)]$, $p_1, ..., p_n > 0$,

$p_1 + ... + p_n = 1$,

$\rho_E \equiv \sum_{a=1}^n p_a \mid \psi_a\rangle\langle\psi_a \mid$, $\langle \hat{Q} \rangle_E = \text{tr}\,(Q_{\rho_E})$. General ρ is positive semi-definite, and $\mathbf{tr}(\rho) = 1$.

Pure state $\longleftrightarrow tr(\rho^2) = 1$.

Spin-one-half-density-matrix: $\rho = \frac{1}{2}(\mathbf{1} + \mathbf{a} \cdot \sigma)$,

$\mid \mathbf{a} \mid \leq 1$.

• (43) Time evolution:

$i\hbar \frac{\partial \rho}{\partial t} = [\hat{H}, \rho]$.

• (44) Schmidt decomposition:

$\mid \psi AB\rangle = \sum_{k=1}^r \sqrt{p_k} \mid \mathbf{k}_A\rangle \otimes \mid k_B\rangle$,

$r \leq \mathbf{d}_A \leq \mathbf{d}_B$,

$\rho_A = \sum_{k=1}^r p_k \mid k_A\rangle\langle k_A \mid$,

$\rho_B = \sum_{k=1}^r p_k \mid k_B\rangle\langle \mathbf{k}_B \mid$,

$\langle k_A \mid k'_A\rangle = \delta_{k,k'}$,

$\langle k_B \mid k'_B\rangle = \delta_{k,k'}$

• (45) Lindblad equation:

$$\frac{\partial \rho}{\partial t} = \frac{1}{i\hbar}[\mathbf{H}, \rho] + \sum_k (\mathbf{L}_k \rho \mathbf{L}_k^\dagger - \frac{1}{2}[\mathbf{L}_k^\dagger \mathbf{L}_k, \rho]) \tag{4.13}$$

• (46) electromagnetic couplings:

$\hat{H}$ = $\frac{1}{2m}(\hat{\mathbf{p}}) - \frac{q}{c}$ $\mathbf{A}$ $(\hat{x}, t)^2$ + q $\phi(\hat{\mathbf{x}, t})$ (no spin).

• (47) Go to equation (1.7)   Gauge-transformations:

$\mathbf{A}'$ = $\mathbf{A} + \nabla \wedge$,

ϕ' = ϕ - $\frac{1}{c}$ $\frac{\partial \wedge}{\partial \mathbf{t}}$,

Ψ' = $\exp(i \frac{q \wedge}{\hbar c})$ Ψ.

• (48) Pauli Hamiltonian (electron):

$\hat{\mathbf{H}}_{Pauli}$ = $\frac{1}{2m_e}(\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A})^2$ + $\frac{e\hbar}{2m_e c}$ $\sigma \cdot \mathbf{B}$ - e ϕ $(\hat{x}, t)$.

• (49) Time-independent Perturbation: non-degenerate:

$\mid n\rangle_\lambda$ = $\mid n^{(0)}\rangle - \lambda \sum_{k \neq n} \frac{\delta \mathbf{H}_{kn}}{\mathbf{E}_k^{(0)} - \mathbf{E}_n^{(0)}} \mid k^{(0)}\rangle$ + $\mathcal{O}(\lambda^2)$,

$E_n(\lambda)$ = $E_n^{(0)}$ + $\lambda \delta \mathbf{H}_{nm}$ - $\lambda^2 \sum_{k \neq n} \frac{|\delta \mathbf{H}_{kn}|^2}{\mathbf{E}_k^{(0)} - \mathbf{E}_n^{(0)}}$ + $\mathcal{O}(\lambda^3)$.

• (50) Degeneracy lifted at $\mathcal{O}(\lambda)$: good-basis:

$\delta \mathbf{H}_{\rangle|} \equiv \langle \Psi_{\mathcal{I}}^{(0)} \mid \delta \mathbf{H} \mid \Psi_{\mathcal{J}}^{(0)} \rangle$ = $\mathbf{E}_{\backslash \mathcal{J}}^{(1)}$ $\delta_{\mathcal{I}\mathcal{J}}$,

$\mid \Psi_{\mathcal{I}}\rangle_\lambda$ = $\mid \Psi_{\mathcal{I}}^{(0)}\rangle - \lambda \sum_p \frac{\delta \mathbf{H}_{p\mathcal{I}}}{\mathbf{E}_p^{(0)} - \mathbf{E}_n^{(0)}} \mid \mathbf{p}^{(0)}\rangle + \sum_{K \neq \mathcal{I}} \frac{|\Psi_K^{(0)}\rangle}{\mathbf{E}_{n\mathcal{I}}^{(1)} - \mathbf{E}_{n\mathcal{K}}^{(0)}} \sum_p \frac{\delta \mathbf{H}_{\mathcal{K}} p \delta \mathbf{H}_p l}{\mathbf{E}(0)_p - \mathbf{E}(0)_n}$ + $\mathcal{O}\lambda^2$,

$\mathbf{E}_{n\mathcal{I}}(\lambda)$ = $\mathbf{E}_n^{(0)} + \lambda \delta \mathbf{H}_{\mathcal{I}I} - \lambda^2 \sum_p \frac{|\delta \mathbf{H}_p \mathcal{I}|^2}{\mathbf{E}_p^{(0)} - \mathbf{E}_n^{(0)}}$ + $\mathcal{O}(\lambda^3)$.

• (51) WKB-quantization with (a, b) turning-points or hard walls:

$\int_a^b k(x')dx$ = $(\mathbf{n} + \beta)\pi$,

$\mathbf{n}$ = $0, 1, 2, ...,$

β = $1 - \frac{1}{4}$ (Number of turning points).

• (52) Time-dependent perturbations:
$\hat{\mathbf{H}}(t)$ = $\hat{\mathbf{H}^{(0)}} + \delta \mathbf{H}(t)$, $|\Psi(t)\rangle \equiv e^{i\hat{\mathbf{H}}^{(0)} \frac{t}{\hbar}} \mid \Psi(t)\rangle$,

$i\hbar \frac{\mathbf{d}}{\mathbf{dt}} \mid \tilde{\Psi}(t)\rangle$ = $\tilde{\delta} \mathbf{H}(t) \mid \Psi(t)\rangle$,

$\delta \mathbf{H}(t) \equiv e^{i\hat{\mathbf{H}}^{(0)} \frac{t}{\hbar}} \delta \mathbf{H}(t) e^{-i\hat{\mathbf{H}^{(0)}} \frac{t}{\hbar}}$,

$\mid \tilde{\Psi}(t)\rangle$ = $\mid \Psi(0)\rangle + \int_0^t \frac{\delta \mathbf{H}(t')}{i\hbar} \mid \Psi(0)\rangle \mathbf{d}t' + \mathcal{O}(\delta \mathbf{H}^2)$

• (53) Fermis'-golden-rule:

$\hat{\mathbf{H}} = \hat{\mathbf{H}^{(0)}} + \mathrm{V}$,

$\hat{\mathbf{H}} = \hat{\mathbf{H}^{(0)}} + 2\mathbf{H}' \cos \omega t$,

$w = \frac{2\pi}{\hbar} \mid \mathbf{V}_f \mid^2 \rho(\mathbf{E}_f)$,

$w = \frac{2\pi}{\hbar} \mid \mathbf{H}'_f \mid 2\rho(\mathbf{E}_f)$

• (54) Adiabatic-approximation:

$\mid \Psi(t)\rangle \simeq e^{i\theta_k(t)} e^{i\gamma_k(t)} \mid \Psi_k(t)\rangle$,

$\hat{\mathbf{H}}(t) \mid \Psi_k(t)\rangle = \mathbf{E}_k(t) \mid \Psi_k(t)\rangle$,

$\theta_k(t) = -\frac{1}{\hbar} \int_0^t \mathbf{E}_k(\mathbf{t}')\mathbf{d}t'$,

$\gamma_k(t) = \int_0^t \nu_k(t')\mathbf{d}t'$,

$\nu_k(t) = i\langle \Psi_k(\mathbf{t}) \mid \dot{\Psi}_k(t)\rangle$.

• (55) Berrys' phase:

$\gamma_n(\Gamma_{if}) = \int_{\Gamma if} \mathbf{A}_n(\mathbf{R}) \cdot d\mathbf{R}$, $\mathbf{A}_n(\mathbf{R}) \equiv i\langle \Psi_n(\mathbf{R}) \mid \nabla_{\mathbf{R}} \mid \Psi_n(\mathbf{R})\rangle$.

• (56) Scattering-on-the-half-line:

$\Psi(\mathbf{x}) = e^{1\delta} \sin(\mathbf{kx} + \delta)$,

$\Psi_s(\mathbf{x}) = e^{i\delta} \sin \delta e^{ikx}$,

$\mathbf{x} > \mathbf{R}$

• (57) Time-delay:

$\triangle\ t = 2\hbar\delta'(\mathbf{E}_0)$.

• (58) Levinsons' Theorem:

$N_b = \frac{1}{\pi}((\delta)(0) - \delta(\infty))$.

• (59) Resonance, Breit-Wigner:

$\mid \Psi_s \mid^2 \simeq \frac{\frac{1}{4}\Gamma^2}{(\mathbf{E}-\mathbf{E}_\alpha)^2 + \frac{1}{4}\Gamma^2}$.

Energy dependence of the squared scattering amplitude $\mid \Psi_s \mid^2$ near resonance.

• (60) Scattering in 3D:

$$\Psi(\mathbf{r}) = \varphi(\mathbf{r}) + \Psi_s(\mathbf{r}) \simeq e^{ikz} + f_k(\theta,\phi)\frac{e^{ikr}}{\mathbf{r}}, \mathbf{r} \gg a. \tag{4.14}$$

$\frac{d\sigma}{d\Omega}$ = $\mid f_k(\theta,\phi) \mid^2$,

σ = $\int \mid f_k(\theta,\phi) \mid^2 \, d\Omega$.

• (61) Rayleigh:

e^{ikz} = $\sqrt{4}\pi \sum_{\ell=0}^{\infty} \sqrt{2\ell+1}\, i^{\ell} \mathbf{Y}_{\ell 0}(\theta) j_{\ell}(kr)$.

• (62) Phase shifts:

$f_k(\theta)$ = $\frac{\sqrt{(4\pi)}}{\mathbf{k}} \sum_{l=0}^{\infty} \sqrt{2l+1} \mathbf{Y}_{l0}(\theta) e^{i\delta_l} \sin\delta_l$,

σ = $\frac{4\pi}{\mathbf{k}^2} \sum_{l=0}^{\infty} (2l+1)\sin^2\delta_l$.

$\Psi\mathbf{r} \mid_l$ = $\mathbf{A}_l \mathbf{j}_l(kr) + \mathbf{B}_l \mathbf{n}_l \mathbf{k}\mathbf{r} \mathbf{Y}_{l0}(\theta)$,

$\mathbf{r} > \mathbf{a}$,

$\tan\delta_l \equiv$ - $\frac{\mathbf{B}_{(l)}}{\mathbf{A}_{(l)}}$.

Chapter 5 Sample practice

5.1 More Useful equations

- d'Alemberts' Principle $\longrightarrow \sum_i \langle \mathbf{F}_i(\mathbf{c}(t), t) - \mathbf{m}_i c_i(t), \mathbf{v}_i \rangle$ = 0 $\forall$ tangent vectors $\mathbf{v}$ at $\mathbf{M}_{\mathbf{c}(t)}$

- n^{th} Taylor Polynomial about x = x_0 for f,

$p_n(x)$ = $f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + ... \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$,
$\sum_{k=0}^{\infty} \frac{(-i*\frac{\theta}{2}*X)^k}{k!}$

- Airy equation $\longrightarrow \frac{d^2\psi}{dU^2}$ - $u\psi(u) = 0$

- Jacobi identity $\longrightarrow [X,[Y,Z]] + [Y,[X,Z]] + [Z,[X,Y]] = 0$

- Stokes theorem(Line-Integrals)

$\iint_s curl\mathbf{F} \cdot d\mathbf{S}$ = $\int_c \mathbf{F} \cdot d\mathbf{r}$, where C is a closed curve, and $\mathbf{F}$ is a vector field defined on C.

- A sympletic-manifold (N, ω), is quantizable (for a particular value of $\hbar$), if $\frac{1}{2\pi\hbar} \int_s \omega \in \mathbb{Z}$, for every closed surface S in N.

- Probability-density-function(Gaussian-distribution):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{5.1}$$

- Poisson bracket of f and g; Moyal bracket $\longrightarrow$ [f,g] =

$\frac{\partial f}{\partial q_1} \frac{\partial g}{\partial p_1}$ +...+ $\frac{\partial f}{\partial q_n} \frac{\partial g}{\partial p_n}$ - $\frac{\partial f}{\partial p_1} \frac{\partial g}{\partial q_1}$ - ... - $\frac{\partial f}{\partial p_n} \frac{\partial g}{\partial q_n}$

- Laguerre Polynomials $\longrightarrow$

$f_{n,l}(r) := L^{2l+1}_{n-l-1}\left(\frac{2r}{nr_B}\right)$, where r_B is the Borh-radius.

- Rydberg formula $\longrightarrow$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{m^2}\right) \tag{5.2}$$

where R_H is the Rydberg constant.

- Rayleigh formula $\longrightarrow$

$\frac{8\pi\nu^2 kT\Delta\nu}{c^3}\ \hbar\, e^{-(\frac{\hbar}{k})(\frac{\nu}{T})}$,

Wiens' formula $\longrightarrow$

$\frac{A\nu^3 e^{-b\nu}}{T\Delta\nu}$, where A and b are constants(does not fit experimental facts for small $\frac{\nu}{T}$, although it avoids the paradox of infinite total energy.

- Radon-Nikodym Theorem(derivative of ν w.r.t μ) $\longrightarrow$

$\nu(E) = \int_E \rho(s)d\mu(s)$
There exists a measurabe function ρ which is the density of ν w.r.t μ

- Schr$\ddot{o}$dinger operator $\longrightarrow$

$\lim_{X\to\infty} \frac{\beta([0,X])}{\beta_q([0,X])} = (2\pi\hbar)^N$, where

$\beta([0, X])$ is a continuous measure space of the set of classic Hamiltonian, $\beta_q([0, X])$ is the discrete measure of the number of eigenvalues < (X), and (N) is the number of dimensions in configuration space.

- Schurs' Theorem $\longrightarrow U^*\mathbf{A}U$ = T = $[T_{ij}] \in M_n(\mathbb{C})$, T_{ij} = 0(i > j).

- Angular momentum in the tensor product space $\longrightarrow$

$\hat{j}_i = \hat{j}_i^{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes \hat{j}_i^{(2)}$ satisfy $[\hat{j}_i, \hat{j}_j] = i\hbar\epsilon_{ijk}\hat{j}_k$ acting on $V_1 \otimes V_2$

- Maxwells'-equations $\longrightarrow$

$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E} = -\frac{\partial B}{\partial t}$
$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right)$

- Abelian/Commutatitive $\longrightarrow \forall$ a,b $\in$ G, a $\cdot$ b = b $\cdot$ a

- Matrix representation of 2-D Hilbert space $\longrightarrow$

$H^2 = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ = I,

where H is for the Hadamard gate, and (I) is the identity.

- Liebniz-Rule $\longrightarrow$

$$\frac{d}{dt}\int_{-\infty}^{\infty} \mid \psi x, t \mid^2 dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} \mid \psi x, t \mid^2 dx = 0 \tag{5.3}$$

- Wave function for a particle in a box: $\longrightarrow$

$$\Psi(x,t) = \sum_{n=1}^{\infty} C_n \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x) e^{-i\left(\frac{n^2\pi^2\hbar}{2ma^2}\right)t} \tag{5.4}$$

- Ground-state for a particle in a box $\longrightarrow \Psi_n$(x) = $\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$

- Gaussian-integral $\longrightarrow$

If $z^2 = ax^2$, and dz = $\sqrt{a}$dx, then $\int_{-\infty}^{\infty} e^{-ax^2}$ dx = $\sqrt{\frac{\pi}{a}}$[Highly simplified]

- **Hookes' Law** $\longrightarrow$

F = - $\frac{dV}{dx}$ = - $\frac{d}{dx}$ $\left(\frac{1}{2}kx^2\right)$ = -k x = m a = m $\frac{d^2(x)}{dt^2}$

• Energy-spectrum of the Transmon qubit $\longrightarrow$

ω_j = $\left(\omega - \frac{\delta}{2}\right) j$ + $\frac{\delta}{2}j^2$

• Pure density operator, and state $\longrightarrow$

$$\mathbf{tr}\rho^2 = tr\rho = \mathbf{1}; \frac{1}{N} < \mathbf{tr}\rho < \mathbf{1} \tag{5.5}$$

$\rightarrow$

positive-definiteness, where $\lambda_j \geq 0$ (λ_j are the eigenvalues, and 'N' is the Hilbert space dimension.

• Traceless-Pauli-matrices $\longrightarrow$

ρ = $\frac{1}{2}$ $\left[\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} + x\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} + y\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} + z\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right]$ = $\frac{1}{2}$ ($\mathbf{I}$ + $\vec{r}\cdot\vec{\sigma}$), where $\vec{r} \in \mathbb{R}^3$, and $\vec{\sigma}$ = (x,y,z).

5.1.1 Test your comprehension

Every equation has a meaning, and can be logically explained.

5.1.1.0.1 Test Case A

Q1. Time-dependent Schr$\ddot{o}$dinger Equation:

Describes how a quantum system evolves $\longrightarrow$

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right)\psi(x,t) = \hat{H}\psi \tag{5.6}$$

• ψ(x,t): The above equation is a wave-function (Probability amplitude).

One-Dimensional Time-independent Schr$\ddot{o}$dinger Equation $\longrightarrow$

-$\frac{i\hbar^2}{2m}$ $\frac{d^2\psi}{dX^2}$ + Vψ = $\hat{H}\psi$

Q2. Heisenberg Uncertainty Principle/Minimum uncertainty wave-packet $\longrightarrow$

$\triangle\ x\ \triangle\ p \geq \frac{\hbar}{2}$(Note that $|\frac{i\hbar\langle\psi|\psi\rangle}{2i}\ |^2$ = $\sqrt{\frac{\hbar^2}{4}}$ = $\frac{\hbar}{2}$).

• You cannot measure Position and Momentum precisely at the same time

• More precision in one, less precision in the other

Q4. Particle in a box Energies $\longrightarrow$

E_n = $\frac{n^2\pi^2\hbar^2}{2mL^2}$, n = 1,2,3,..

- $\hbar$: reduced Plancks' constant($\frac{\hbar}{2\pi}$) [h is the Plancks constant].

- i: imaginary unit/Lateral-unit

- $\hat{H}$: Hamiltonian (total energy operator)

Q5. Commutation-relation $\longrightarrow$ [$\hat{x}, \hat{p}$] = i$\hbar$

Significance:

- Leads to uncertainty principle

- Shows position and momentum, are incompatible observables

5.1.1.0.2 Test Case B

Q6. Particle in a box

(a) Wave-function

Boundary conditions:

- ψ(0) = 0 $\implies$ $\mathbf{B}$ = 0

- $\psi(\mathbf{L})$ = 0 $\sin(k\mathbf{L})$ = 0

So:

k = $\frac{n\pi}{L}$

Final wave-function:

ψ_n(x) = $\sqrt{\frac{2}{L}}\,\sin\left(\frac{n\pi x}{L}\right)$

(b) Energy-Level

$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$

(c) Probability

$\mathbf{P} = \int_0^{\frac{L}{2}} |\psi_1|^2$ dx = $\frac{2}{L}\int_0^{\frac{L}{2}} \sin^2\left(\frac{\pi x}{L}\right)$ dx

Result:

$\mathbf{P} = \frac{1}{2}$

Q7. Commutator and Uncertainty

(a) Show:

$[\hat{x}, \hat{p}]$ = i$\hbar$

Apply to function f(x):

$\hat{x}\hat{p}$f = x(-i$\hbar$f')

$\hat{p}\hat{x}$f = -i$\hbar$ $\frac{d}{dx}$(xf) = -i$\hbar$(f + xf')

Subtract:

$[\hat{x}, \hat{p}]$f = i$\hbar$f

Thus:

$[\hat{x}, \hat{p}]$ = i$\hbar$

(b) Uncertainty-Principle

General relation:

$\triangle$ $\mathbf{A}$ $\triangle$ $\mathbf{B} \geq \frac{1}{2} \mid \langle[\mathbf{A}, \mathbf{B}]\rangle \mid$

Substitute:

$\triangle$ x $\triangle$ $p \geq \frac{\hbar}{2}$

Q9. Hydrogen-Atom

(a) Energy-Level

E_n = - $\frac{13.6}{n^2}$ eV

(b) Quantum-number

- n: Energy-Level
- ℓ: orbital-shape
- m: orientation

(c) Why Energy is Negative

- Electron is bound to nucleus
- Energy is required to remove electron

• Bound-states

Wave-function-collapse

• Before measurement, we have superposition

• Measurement leads to a definite state

A collapse = Transition from probability to reality$\mathbb{R}$

5.1.1.0.3 Additional Notes

I.(a) Completeness

A Hilbert space is complete if:

Every Cauchy sequence $\psi_n \subset \mathscr{H}$ converges to a limit in $\mathscr{H}$

Formally:

$\|\psi_n - \psi_m\| \longrightarrow 0 \implies \exists \psi \in \mathscr{H}, \psi_n \longrightarrow \psi$

(b) Extension to Orthonormal basis

Given orthonormal-set e_1:

• If not complete, there exists $\psi \neq 0$ orthogonal to all e_1

• Add ψ, normalize enlarge set

Zorns'-Lemma argument:

Maximal orthonormal-set = orthonormal basis

(c) Projection is bounded

Projection-operator $\mathbf{P}$:

$\mathbf{P}^2$ = $\mathbf{P}$, $\mathbf{P}$† = $\mathbf{P}$

For any ψ:

$\|\mathbf{P}\psi\| \leq \|\psi\|$

Thus:

$\|\mathbf{P}\| \leq 1.$ therefore bounded.

II. Spectral-Theorem

(a) Statement

For self-adjoint-operator $\hat{A}$:

$\hat{A} = \int \lambda \, dE(\lambda)$

• $E(\lambda)$: Projection-valued-measure

• Generalizes diagonalization

(b) Spectrum-Types

• $\mathbf{Dicrete}$ eigenvalues (Bound-states)

• $\mathbf{Continuous}$ scattering-states
(c) Measurement-Interpretation

Probability of outcome in set $\triangle$:

$\mathbf{P}(\triangle) = \langle\psi|E(\triangle)|\psi\rangle$

Measurement-outcome = spectrum of operator

III. Operators and Dynamics

Time Evolution

(a) Derivation

Start:

$\langle\mathbf{A}\rangle = \langle\psi|\hat{\mathbf{A}}|\psi\rangle$

Differentiate:

$\frac{d}{dt}\langle\mathbf{A}\rangle = \langle\, \frac{\partial\psi}{\partial t}|\mathbf{A}|\psi\rangle + \langle\psi|\mathbf{A}|\frac{\partial\psi}{\partial t}\rangle + \langle\psi|\frac{\partial\mathbf{A}}{\partial t}|\psi\rangle$

Use Schr$\ddot{o}dinger$ equation:

$i\hbar\frac{\partial\psi}{\partial t} = \mathbf{H}\psi$

Substitute:

$\frac{d\langle\mathbf{A}\rangle}{dt} = \frac{i}{\hbar}\,\langle[\hat{\mathbf{H}}, \hat{\mathbf{A}}]\,\rangle + \langle\frac{\partial\mathbf{A}}{\partial t}\rangle$

(b) Apply to position

$\hat{\mathbf{H}} = \frac{p^2}{2m}$

$\hat{\mathbf{H}}$,x = $\frac{1}{2m}$ $[p^2, x]$

Use:

[$\hat{p}$,x] = -i$\hbar$

Result:

$\frac{d\langle x\rangle}{dt} = \frac{\langle p\rangle}{m}$

Velocity operator emerges

(c) Conservation condition

If:

$[\hat{\mathbf{H}}, \hat{\mathbf{A}}] = 0$ and $\frac{\partial A}{\partial t} = 0$

Then:

$\frac{d\langle A\rangle}{dt} = 0$

A is conserved

IV. Harmonic Oscillator Algebra

(a) Commutator

$\hat{a} = \frac{1}{\sqrt{2\hbar m\omega}}$(m ω x + ip)

Compute:

$[\hat{a}, \hat{a}\dagger] = 1$

(b) Hamiltonian

$$\hat{\mathbf{H}} = \hbar\omega(\hat{a}\dagger\hat{a} + \frac{1}{2}) \tag{5.7}$$

(c) Orthonomality

$\langle n \mid m\rangle = \delta_{nm}$

Proof:

- Ladder-Operators generate states
- Inner-product preserved recursively

V. Advanced Systems

Angular-Momentum

(a) Lie Algebra

$[J_i, J_j] = \mathrm{i}\hbar\epsilon_{ijk}J_k$

Derived from rotational symmetry generators

(b) Ladder-Operators

$J_\pm = J_x \pm \mathrm{i}J_y$

$[J_z, J_\pm] = \pm\hbar J_\pm$

(c) Eigenvalue

$J^2 \mid jm\rangle = \hbar^2 j(j+1)|jm\,\rangle$

$J_z \mid jm\rangle = \hbar m \mid jm\rangle$

VI. Perturbation-Theory

(a) First Order

$E_n^{(1)} = \langle n^{(0)}|\mathrm{V}|n^{(0)}\rangle$

(b) Second Order

$E_n^{(2)} = \sum_{k\neq n} \frac{|\langle k|V|n\rangle|^2}{E_n^{(0)} - E_k^{(0)}}$

(c) Degeneracy

• Degenerate-states mix

• Solve matrix:

$\mathbf{V}_{ij} = \langle i \mid \mathbf{V}|j\rangle$

Diagonalize within subspace

VII. Conceptual

Decoherence

• System interacts with environment

• Phase information lost

• Superposition(classical mixture)

Explains Classical world emergence

VIII. Path-Integral

Core Idea
Amplitude = $\sum_{paths} e^{\frac{i}{\hbar}S[x(t)]}$

Action

S = $\int$L dt

Classical-Limit
• Does not account for a dominant path/stationary action

• Quantum Mechanics recovers Classical Mechanics

Quantum Mechanics becomes Linear Algebra, Operator Theory, and Symmetry, NOT just formulas.

Remark

This appendix covers some of the Mathematics used in this short exposition of operators, and linear equations. Here you will find the special functions that are relevant to our current application. These require algebraic techniques.

Calculus is also necessary to understand much of this material, for it is always used in advanced topics.

Chapter 6 Example Functions

6.0.1 Equations

Plane-waves $\longrightarrow$

$\psi(x,t)$ = $e^{(ikx-i\omega t)}$; we then differentiate this function to obtain the Energy, and Momentum operator, using the Chain Rule...and applying Plancks' rule(multiplying by i$\hbar$).

$i\hbar\frac{\partial}{\partial t}\psi(x,t)$ = $i\omega e^{(ikx-i\omega t)}$ = $i\omega\psi(x,t)$, seeing the eigenfunction on the left side of the equation(taking the derivative), and the resulting constant eigenvalue on the right hand side of the equation.

Energy-operator($\hat{E}$) $\rightarrow$

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = i\hbar(-i)\omega\psi(x,t) = \hbar\psi(x,t) \tag{6.1}$$

Momentum operator($\hat{P}$) $\rightarrow$

$$(-i\hbar)\frac{\partial}{\partial x}\psi(x,t) = (-i\hbar)ike^{ikx-i\omega t} = \hbar k\psi(x,t) \tag{6.2}$$

(Note -i cancels i), k = $\frac{2\pi}{\lambda}$; ω = kc

Vector and Matrix multiplication $\rightarrow$ A$\vec{v}$ = λ $\vec{v}$, where λ is an eigenvalue, and $\vec{v}$ is the eigenvector.

Energy of a photon $\rightarrow$ E = $\hbar\omega$; $\vec{P}$ = $\hbar$k ; $\triangle$ m = 0(m is for mass); E = Pc

Discrete Fourier Transform $\rightarrow$

Consider the space of 2π periodic functions f: $\mathbb{R} \rightarrow \mathbb{C}$, with a continuous 1st derivative; We have $E_k(x)$ = e^{ikx}, k $\in$ $\mathbb{Z}$. Using Eulers' formula($e^{i\alpha}$ = $\cos\alpha + i\sin\alpha$); $e^{2\pi i}$ = 1, $E_k(x)$by substitution, $\equiv$ $E_k(x)$ = $\cos(kx) + i\sin(kx)$; Here E_k is an orthonormal set.

Now, for the Hermitian Scalar Product $< f \mid \frac{1}{2\pi}\int_0^{2\pi}\overline{f(x)}$ $\cdot$ g(x) dx, we have $< E_k \mid E_s >$ = $\frac{1}{2\pi}\int_0^{2\pi} e^{-ikx} \cdot e^{isx}$ = $\frac{1}{2\pi}\int_0^{2\pi} e^{i(s-k)x}dx$(factored), = $\frac{1}{2\pi}\ \frac{1}{i(s-k)}\ e^{i(s-k)x}\ |_{x=0}^{x=2\pi}$.

Note that $\iff$ s and k = 0, $\begin{cases} 0, \text{if s} \neq \text{k} \\ 1, \text{if s = k} \end{cases}$

Since its periodic($\frac{1}{2\pi}\int_0^{2\pi}$1 dx = 1, if s = k); With the resulting anti-derivative of the function. So, we can pass from $(f_0, f_1, ..., f_{n-1})$ $\rightarrow$ Fourier coefficients $(c_0, c_1, ..., c_{n-1})$. This is the Discrete Fourier series; A discrete signal capturing frequency information, given that C_k = $\frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} f_j\ e^{\frac{i2\pi kj}{N}}$(DFT); Note that C_k = $< E_k \mid f >$ = $\sum_{j=0}^{N-1}\overline{E_{kj}}\ f_j$, where N is the number of samples.

Inverse Discrete Fourier Transform(IDFT) $\rightarrow$

f_j = $\sum_{k=0}^{N-1} C_k\ E_{kj}$ = $\frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} C_k\ e^{-\frac{i2\pi kj}{N}}$.

Langranges' Theorem $\rightarrow$

Let G be a group of an order $|G|_n$; Then, $\forall$ a $\in$ G, a^n = e; So, $\mathbb{Z}_p^*$ = G. It has a multiplicative inverse.

Fermats' Little Theorem $\rightarrow$ Let p be a prime; Then $\forall$ a $\in \mathbb{Z}_p^*$, a^{p-1} = 1 mod p, (a) is not divisible by p.

In an RSA setting, m = $p \cdot q$, p and q are Primes; For n = $|\mathbb{Z}_m^*|$ = (p -1) $\cdot$ (q - 1); If (a) is not divisible by p or q, then $a^{(p-1)(q-1)}$ = 1 mod pq.

Hadamard $\longrightarrow$ H; $\mid 0 > \rightarrow$

$\frac{1}{\sqrt{2}} \mid 0 >+\mid 1 >$, and $\mid 1 > \rightarrow -\frac{1}{\sqrt{2}} \mid 0 > + \mid 1 >$

H = $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
Eulers' Identity $\rightarrow e^{i\pi}$ = -1

Adiabatic-approximation $\rightarrow \Psi(t)$ = $e^i\theta_k(t)$ $e^i\gamma_k(t) \mid \mid \psi_k(t) >$

Lowest-Landau-level $\rightarrow$

$$\Psi_m(z) \propto z^m e^{-|z|^2} \tag{6.3}$$

Remark Quantum Fourier Transform $\rightarrow U_{QFT}$ is not a Hermitian operator, $U_{QFT} \neq U_{QFT}^\dagger$. It creates an equally weighted superposition of states, leaving amplitudes as complex numbers, while Hadamard leves them as $\mathbb{R}$. QFT also allows derivation of Fourier Basis, used in Quantum Mechanics inorder to ease and describe various calculations involved in the Theory.

Baker-Campbell-Hausedorff Theorem $\rightarrow$

$$\log(e^X e^Y) = X + \int_0^1 g(e^{\text{ad X}} e^{\text{t ad Y}})(Y)dt. \tag{6.4}$$

For a complete derivation, see Dynkins' analysis of Lie groups.
Perturbed-System $\rightarrow$

$\hat{H}$ = $\frac{\hat{p}^2}{2m}$ + $\frac{1}{2}m^2\omega^2x^2$ - $\frac{\lambda}{2}x^2$, with an exact solution of E $\approx \frac{\hbar}{2}$ - $\frac{\hbar\lambda}{4m\omega}$

Perturbation-Theory $\rightarrow$

$$E = E^{(0)} + \langle\psi^{(0)} \mid \lambda\hat{w} \mid \psi^{(0)}\rangle = \frac{\hbar\omega}{2} + \langle\psi^{(0)} \mid -(\frac{\lambda}{2})x^2 \mid \psi^{(0)}\rangle = \frac{\hbar\omega}{2} - \lambda\frac{\lambda\hbar}{4m\omega} = -\frac{\lambda}{2}\langle x^2\rangle = -\frac{\lambda}{2}\sigma^2. \tag{6.5}$$

Note that $\psi^{(0)}$(x) = c$e^{-\frac{x^2}{4\sigma^2}}$ and σ^2 = $\frac{\hbar}{2m\omega}$.

Cross-Product $\rightarrow$

The cross product of u = (u_1, u_2, u_3) and v = (v_1, v_2, v_3) is a vector (u $\times$ v) =

$\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$ = $(u_2v_3 - u_3v_2)\mathbf{i}$ + $(u_3v_1 - v_1u_3)\mathbf{j}$ + $(u_1v_2 - u_2v_1)\mathbf{k}$.

The vector (u $\times$ v) is perpendicular to u and v.

The cross product (v $\times$ u) is -(u $\times$ v).

Note that (u $\times$ u) = 0,

$\|u \times v\|$ = $\|u\|$ $\|v\|$ | $\sin\theta$ |, and | $u \cdot v$ | = $\|u\|$ $\|v\|$ | $\cos\theta$ |.

Time evolution Hamiltonian operator, form a complete basis $\rightarrow$

$$i\frac{d}{dt} \mid \psi\rangle(t) = i\frac{d}{dt}\hat{U}(t) \mid \psi(0)\rangle = \hat{H}\hat{U}(t) \mid \psi(0)\rangle = \hat{H} \mid \psi(t)\rangle \tag{6.6}$$

Note that $\frac{d}{dt}$ $\hat{U}$(t) = $-\frac{i}{\hbar}$ $\hat{H}$ $\hat{U}$(t), its like the Schr$\ddot{o}$inger equation!, and $\hat{U}$(t)$|\psi_n\rangle$ = $e^{\frac{-iE_n t}{\hbar}}$ $\psi_n\rangle$ too.

Quantum-jump $\rightarrow$

An abrupt projection of a system to an Eigenstate ψ_0, with a dissipative process(spontaneous emission).

The evolution of the system starts from zero, ruled by the effective Hamiltonian. The modification of $\psi(t)$ by **non-observation** of spontaneous emission, reduces the population of the state excited by 1 - $\frac{1}{2}\Gamma$dt, while the ground-state remains unchanged. Every **quantum jump** projecting the system into the ground-state constitutes a **measurement**.

Monte Carlo Wave-function(MCWF) simulation $\rightarrow$

ψ(t) $\hookrightarrow$ | $\psi(t + dt)\rangle \equiv$

$$\left\{ \begin{array}{l} \frac{(1-i\hat{H}dt)|\psi(t+dt)\rangle}{\sqrt{\langle\psi(t)|\psi(t)\rangle}} \text{ if } \zeta > 1 - \langle\psi(t) \mid \psi(t)\rangle \\ \mid \psi_0\rangle \text{ if } \zeta < 1 - \langle\psi(t) \mid \psi(t)\rangle \end{array} \right\}.$$

Concept of Magnitude $\rightarrow$ If w = 3 + 4i, then | w | = $\sqrt{\Re(w^2) + \Im(w^2)}$ = $\sqrt{3^2 + 4^2}$ = 5

Cube roots of 1 $\rightarrow$

z^3 = 1 $\implies$ $(x + iy)^3$ = 1 $\implies$ x^3 + $3x^2(iy)$ + $3x(iy)^2$ + $(iy)^3$ = 1 $\implies$ $(x^3 - 3xy^2)$ + $i(3x^2y - y^3)$ = 1;

So, $x^3 - 3xy^2$ = 1, and $3x^2y - y^3$ = 0.

x = 1, y = 0, and z = 1, is a solution.

If x and y are allowed to be complex, then there are three solutions (if x is non-zero). It has three cube roots. A real number has one real cube root, and two further cube roots which form a complex conjugate pair. For instance, the cube roots of 1 are: 1, $\frac{1}{2}$ + $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$ - $\frac{\sqrt{3}}{2}$.

Note that,
the principal cube root of a negative number is a complex number, instance $\sqrt[3]{-8}$ is NOT -2, but rather (1 + i$\sqrt{3}$). Quartic equations can also be solved in terms of cube roots and square roots.

Seiberg-Witten functional $\rightarrow$ $D_{A\varphi}$ = 0,

$$F_A^+ = \frac{1}{4}\langle e_j \cdot e_k \cdot \varphi, \varphi\rangle e^j \wedge e^k \tag{6.7}$$

Fundamental electron velocity of the Hydrogen states $\rightarrow$

$$E_1 = \frac{-\hbar^2}{2m_e a_B^2} \tag{6.8}$$

Relativistic velocity of the electron $\rightarrow$

v = $\sqrt{\frac{2E_1}{m_e}}$ = $\frac{\hbar}{m_e a_B}$ = $\frac{e^2}{4\pi\epsilon_0\hbar}$ = αc

Velocity-operator $\langle\hat{v}\rangle \rightarrow$

$\frac{i\hbar}{m} \int \psi^* \frac{\partial\psi}{\partial x}$ dx

[Velocity as motion of expectation, applies to Momentum(p), $\mathbf{H}$(Hamiltonian total Energy of a system), and Kinetic Energy + Potential Energy (T).

Klein-Gordon equation $\rightarrow$

$\frac{1}{c^2} \frac{\partial^2}{\partial t^2}$ - ∇^2 + $(\frac{m_e c}{\hbar})^2$] ψ = 0

Note that $[\partial^\mu, \partial^n u$ = $[\partial^{\mu\nu}]$. The matching-index-height rule is satisfied.

6.1 Sampled wave-function distribution

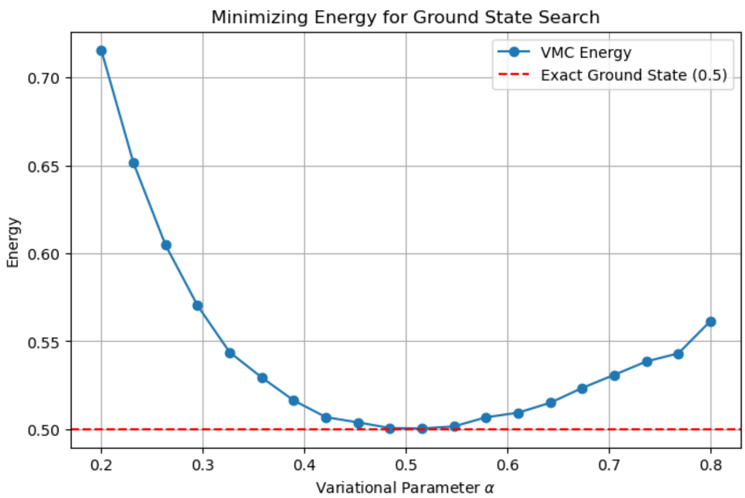

Figure 6.1: Mnimizing energy for a ground-state

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[**https://citizendium.org/wiki/index.php?title=Particle_in_a_box&oldid=984214**]

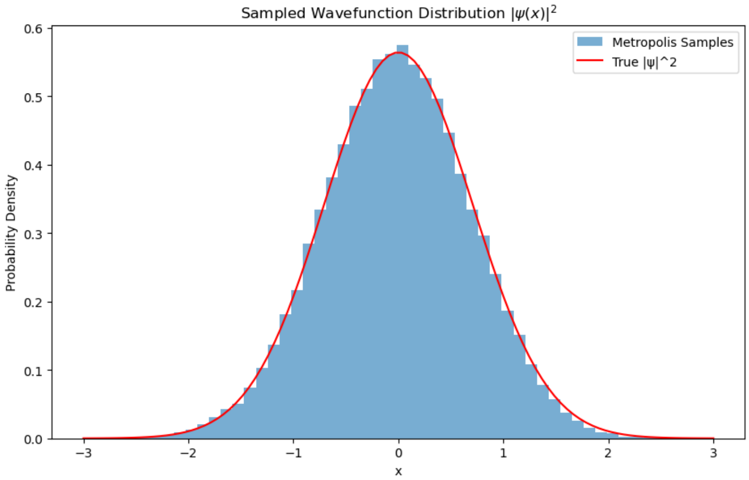

Figure 6.2: Estimated Ground State Energy, with its' analytical solution

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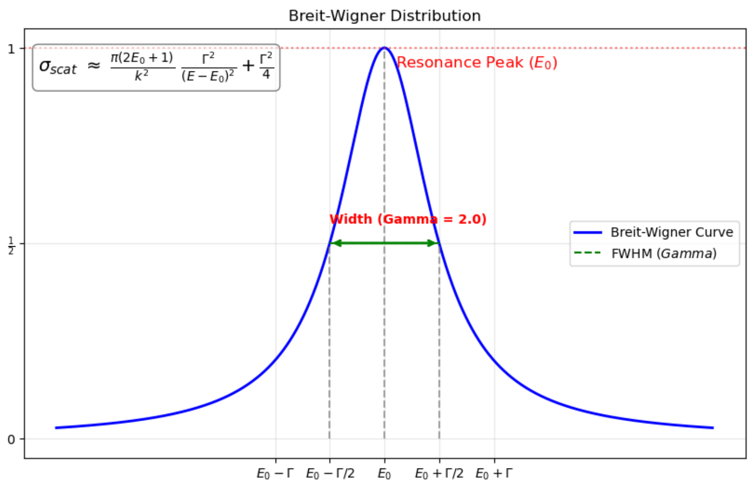

Figure 6.3: Breit-Wigner distribution

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Chapter 7 Appendix

7.1 Rules for Continuous Functions[Apply Limit Laws]

Product Rule $\rightarrow$ Let f(x) and g(x) be differential functions. Then,

$$\boxed{\frac{d}{dx}f(x)g(x) = \frac{d}{dx}[f(x)\cdot g(x)] + \frac{d}{dx}[g(x)\cdot f(x)]} \tag{7.1}$$

That is, if [f(x) = f(x) g(x)], then [f(x) = $f^{'}(x)g(x) + g^{'}(x)f(x)$].

This means that the derivative of a product of two functions is the derivative of the first function times the second function, plus, the derivative of the second function times the first function.

1. **Quotient Rule**
2. **Power Rule**

Chain Rule $\rightarrow$ Let f and g be functions. $\forall$ x in the domain of g, for which g is differentiable at x, and f is differentiable at g(x), the derivative of the composite function h(x) = f(g(x) is given by

$$h'(x) = f'(g(x)g'(x) \tag{7.2}$$

Alternatively, if (y) is a function of (u), and (u) is a function of (x), then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ e.g $\frac{d}{dx}(2x^2-5)^5 = 20(x)(2x^2-5)^4$

Contour Integrals $\rightarrow$ If the function f(z) has isolated singularities (poles) inside a closed contour (γ), the integral around that contour is equal to 2πi times the sum of the residues at those singularities; $\oint_\gamma f(z)dz = 2\,\pi\mathrm{i}\sum \mathrm{Res}\,(f, z_k)$

7.2 List of experiments

1. Davisson-Germer
2. Photo-electric effect
3. Youngs double-slit
4. Stern-Gerlach
5. EPR
6. Badurek-Rauch-Tuppinger
7. SQUID(A Superconducting Quantum Interference Device (SQUID) experiment, measures extremely weak magnetic fields by exploiting Josephson junctions, and magnetic flux quantization within a superconducting loop, cooled by liquid nitrogen at 77°K or -196.1°C.
8. Delayed-choice

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